

# Cboe Titanium
# Cboe Australia Equities
# Binary Order Entry Version 3 Specification

Version 1.0.20

22 May 2026

This content is owned or licensed by Cboe Global Markets, Inc. or its affiliates (“Cboe”) and protected by copyright under U.S. and international copyright laws. Other than for internal business purposes, you may not copy, reproduce, distribute, publish, display, perform, modify, create derivative works, transmit, or in any way exploit the content, sell or offer it for sale, use the content to construct any kind of database, or alter or remove any copyright or other notice from copies of the content.

# Contents

**1. Introduction ..... 4**
1.1 Overview ..... 4
1.2 BOEv3 Message Format Versioning and Nomenclature ..... 4
1.2.1 Introduction of New Fields in Existing BOEv3 Messages ..... 5
1.3 Hours of Operation ..... 5
1.4 Data Types ..... 5
1.5 Protocol Features ..... 6
1.5.1 Architecture ..... 6
1.5.2 Maximum Open Order Limits ..... 6
1.5.3 Stale NBBO ..... 6
1.5.4 Port Types ..... 6

**2. Session ..... 8**
2.1 Message Headers ..... 8
2.1.1 Handling of Invalid Message Headers ..... 8
2.2 Login, Replay, and Sequencing ..... 8
2.3 Sequence Reset ..... 9
2.4 Heartbeats ..... 10
2.5 Logging Out ..... 10

**3. Session Messages ..... 11**
3.1 Participant to Cboe Australia ..... 11
3.1.1 Login Request ..... 11
3.1.2 Logout Request ..... 12
3.1.3 Example Logout Request Message ..... 13
3.1.4 Client Heartbeat ..... 13
3.2 Cboe Australia to Participant ..... 14
3.2.1 Login Response ..... 14
3.2.2 Replay Complete ..... 15
3.2.3 Logout Response ..... 16
3.2.4 Server Heartbeat ..... 17

**4. Application Messages ..... 18**
4.1 Participant to Cboe Australia ..... 18
4.1.1 New Order ..... 18
4.1.2 Modify Order ..... 28
4.1.3 Cancel Order ..... 34
4.1.4 Purge Orders ..... 34
4.2 Cboe Australia to Participant ..... 36
4.2.1 Order Acknowledgement ..... 36
4.2.2 Order Rejected ..... 38
4.2.3 Order Modified ..... 39
4.2.4 Modify Rejected ..... 42
4.2.5 Order Restated ..... 43
4.2.6 Order Execution ..... 45
4.2.7 Order Cancelled ..... 47
4.2.8 Cancel Rejected ..... 48
4.2.9 Order Execution Restated ..... 48
4.2.10 Purge Acknowledgement ..... 49

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

4.2.11 Purge Rejected ..... 50
4.2.12 Trade Cancel or Correct ..... 51
4.2.13 Carried Restatement ..... 52
4.2.14 Done For Day ..... 53
4.3 Application Message Fields ..... 54
4.4 Order Reason Codes ..... 65
**5. Port Attributes ..... 67**
**6. Support ..... 69**
**Appendix A – Login Strategies ..... 70**
Login Message Flow ..... 70
Recommended Login ..... 70
Other Login Strategies ..... 71
**Revision History ..... 73**

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

# 1. Introduction

## 1.1 Overview

This document describes Binary Order Entry, version 3 (BOEv3), the Cboe Australia proprietary order entry protocol used by a participant to send orders to Cboe Australia.

Where applicable, the terminology (e.g., time in force) used in this document is similar to that used by the FIX protocol to allow those familiar with FIX to more easily understand BOEv3. This document assumes the reader has basic knowledge of the FIX protocol.

BOEv3 fulfills the following requirements:

- *CPU and memory efficiency.* Message encoding, decoding, and parsing are simpler to code and can be optimised to use less CPU and memory at runtime.
- *Application level simplicity.* State transitions are simple and unambiguous. They are easy to apply to a participant's representation of an order.
- *Session level simplicity.* The session level protocol (login, sequencing, replay of missed messages, logout) is simple to understand.

While Cboe Australia has striven to preserve feature parity between FIX and BOEv3 where possible, some features may only be available in one protocol or the other.

Note that while FIX is an ASCII based protocol, BOE is binary based, providing for efficiencies that can allow for reduced latency.

All binary values are in **little-endian** (as used by Intel x86 processors), and not network byte order.

Each message is identified by a unique message type. A listing of the supported message types is provided in Table 1 - Session Message Types and Table 2 - Application Message Types.

All communication is via standard TCP/IP.

## 1.2 BOEv3 Message Format Versioning and Nomenclature

Since BOEv3 uses a more rigidly structured message format than prior versions of BOE, message types may be introduced when new fields cannot be accommodated by utilising reserve bytes in the existing message specification. In such cases, when a new message type is introduced, it will be documented as a distinct message type in this document. Application layer message types are named using the following pattern:

| Application Layer Message Type | General Type | Market | Version |
|---|---|---|---|
| NewOrderAUEquitiesV1 | New Order | AU Equities | V1 |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

This allows for ease of distinction between similar message types between markets (for example, AU Equities compared to US Futures), and the handling of new versions of the message (V1, V2, *et cetera*).

When application layer message types are discussed in this document, they are being referred to using their general type name unless their specific version is relevant to the documentation.

During any time when multiple versions of messages from Cboe Australia to Participant are supported concurrently, the configuration of the port determines which message version may be sent by Cboe Australia.

### 1.2.1 Introduction of New Fields in Existing BOEv3 Messages

Existing message fields will not change in length. Fields currently identified as "Reserved" may be redefined, in part or in whole, as new specified fields. New fields may be introduced at the end of any message that does not have a repeating group of fields; consequently, Clients must check the length of messages received from Cboe Australia and treat any additional bytes present as undefined values.

## 1.3 Hours of Operation

Refer to the website for the Cboe Australia Regular Trading Hours and Cboe Australia Holiday Schedule.

## 1.4 Data Types

The following data types are used by BOEv3. The size of some data types varies by message. All data types have default values of binary zero, in both Participant to Cboe Australia and Cboe Australia to Participant contexts.

- *Binary:* Little Endian byte order, unsigned binary value. The number of bytes used depends on the context.
    - One byte: FE = 254
    - Four bytes: 64 00 00 00 = 100
- *Binary Price:* Little Endian byte order value, signed two's complement, eight bytes in size, with four implied decimal places. So, if the value is 123,400, the actual value considering implied decimal places is 12.34.
    - 08 E2 01 00 00 00 00 00 = 123,400/10,000 = 12.34
    - F8 1D FE FF FF FF FF FF = -123,400/10,000 = -12.34
- *Alpha:* ASCII uppercase letters (A-Z) and lowercase letters (a-z) only. ASCII NUL (0x00) filled on the right, if necessary. The number of bytes used depends on the context.
- *Alphanumeric:* ASCII uppercase letters (A-Z), lowercase letters (a-z) and numbers (0-9) only. ASCII NUL (0x00) filled on the right, if necessary.
- *Text:* Printable ASCII characters only (binary values in the inclusive range 0x20 through 0x7E). ASCII NUL (0x00) filled on the right, if necessary. The number of bytes used depends on the context.

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

- *DateTime:* Little Endian byte order, unsigned binary value, 8 bytes in size. The date and time, in UTC, represented as nanoseconds past the UNIX epoch (00:00:00 UTC on 1 January 1970). The nanoseconds portion is set to 0 by the exchange (the actual precision will be in microseconds).
    - E0 FE 20 F7 36 71 F8 11F = 1,294,909,373,757,324,000 = 2011-01-13 09:02:53.757324 UTC
- *Date:* Little Endian byte order, unsigned binary value, 4 bytes in size. The YYYYMMDD expressed as an integer.
    - A7 3C 34 01 = 20200615 = "June 15, 2020"
- *Reserved*: sequence of ASCII NUL (0x00) values when sent by the participant. May contain any values when sent by the exchange and should be ignored by the participant.

## 1.5 Protocol Features

The exchange does not guarantee messages sent by Members to the exchange, including through protocols such as TCP. Members are responsible to monitor the status of the messages they send to the exchange.

### 1.5.1 Architecture

Each BOE order handler process will allow a single TCP connection from a participant. Connection attempts from unknown source IP ranges will be blocked to prevent unauthorised access to BOE ports. Contact the Cboe APAC Network Operations Centre to connect from a new source IP range.

Each BOE order handler will connect, using a proprietary UDP protocol, to all matching units. Connections from order handlers to matching engines are latency equalised. The connections between order handlers and matching units are governed by an internal flow control mechanism to control burst rates.

### 1.5.2 Maximum Open Order Limits

The exchange limits the maximum number of open orders allowed on a BOE port to 300,000 per port. New orders will be rejected once this limit is breached until the number of open orders drops back below the limit.

### 1.5.3 Stale NBBO

A stale ASX BBO will occur when the Cboe Australia trading system determines that one or more market data channels are impaired or down completely. If the trading system detects that an ASX BBO is stale, trading will continue using the Cboe Australia BBO quote for all pricing.

### 1.5.4 Port Types

All BOE port types may be ordered through the Customer Web Portal using the Logical Port Request form tool. Changes to port attributes may be requested through the same tool by submitting a 'Modify' request for one or more existing BOE ports.

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

#### *1.5.4.1 BOE Order Ports*

BOE Order Ports support order entry, modification, and cancellation. Order Ports purposefully do not support the usage of the `Purge Orders` message type. The attempted usage of this message type on a standard BOE order port will result in a rejection of the disallowed message type.

BOE Order Ports are limited to 4,000 inbound messages per second. Once the inbound limit is reached new orders are rejected, modifies are handled as cancels, and cancels are processed normally. BOE Order Ports are limited to 10 inbound messages per second on Cboe Australia Test Products (including ZZA).

#### *1.5.4.2 BOE Matching Units and Sequence Numbers*

BOE Order Ports are not specific to a particular matching unit. Note however that message sequence numbers from Cboe Australia are issued per matching unit. Sequence numbers will be monotonically incremental values per *MatchingUnit*. Currently, Cboe Australia has 2 matching units.

#### *1.5.4.3 BOE Purge Ports*

BOE Purge Ports support only one message type from participants: `Purge Orders`. Participants may use this port type to request a cancellation of groups of orders, including orders across multiple BOE/FIX Order Ports.

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

# 2. Session

## 2.1 Message Headers

Each message has a twelve-byte header. The two initial *StartOfMessage* bytes are present to aid in message reassembly for network capture purposes. The *MatchingUnit* field is only populated on sequenced, non-session level messages sent from Cboe Australia to the Participant. Messages from Participant to Cboe Australia and all session level messages must always set this value to 0.

| Field Name | Offset | Length | Data Type | Description |
|---|---|---|---|---|
| StartOfMessage | 0 | 2 | Binary | `B0 E3` (58288) |
| MessageLength | 2 | 2 | Binary | Number of bytes for the message, including this field but not including the two bytes of the *StartOfMessage* field. |
| MessageType | 4 | 2 | Binary | Message type |
| MatchingUnit | 6 | 1 | Binary | Matching unit which created this message. Matching units in BOE correspond to matching units on Multicast PITCH. For session level traffic the unit is set to 0. For messages from Participant to Cboe Australia, the unit must be 0. |
| Reserved | 7 | 1 | Binary | Must be zero from participant. Value unspecified from Cboe Australia. |
| SequenceNumber | 8 | 4 | Binary | The sequence number for this message. Messages from Cboe Australia to Participant are sequenced distinctly per matching unit. Zero for session level traffic. |

### 2.1.1 Handling of Invalid Message Headers

If an invalid message header is encountered, the exchange will disconnect the port. A message header is considered invalid if any of the following is encountered:

- StartOfMessage is not `B0E3`.
- *MessageLength* is not appropriate for the given message type.
- *MessageType* is not a documented message type for Cboe Australia. Note that the types of application messages accepted will vary between Cboe Australia and other BOEv3 exchanges.

## 2.2 Login, Replay, and Sequencing

Session level messages, both inbound (Participant to Cboe Australia) and outbound (Cboe Australia to Participant) are unsequenced.

Inbound (participant to Cboe Australia) application messages are sequenced. Upon reconnection, Cboe Australia informs the participant of the last processed sequence number; the participant may choose to resend any messages with sequence numbers greater than this value. A gap forward in the

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

participant's incoming sequence number is permitted at any time and is ignored by Cboe Australia. Gaps backward in sequence number (including the same sequence number used twice) are never permitted and will always result in a `Logout Response` message being sent and the connection being dropped.

Most (but not all) outbound (Cboe Australia to participant) application messages are monotonically sequenced per matching unit. Each message's documentation will indicate whether it is sequenced or unsequenced. While matching units on BOE correspond directly to matching units on Multicast PITCH, sequence numbers do not.

Upon reconnection, a participant sends the last received sequence number per matching unit in a `Login Request` message. Cboe Australia will then respond with any missed messages.

The *ReplayUnspecifiedUnit* value can be used to control the replay behaviour for unknown units. If the flag is set to F (Fail), Cboe Australia will send a `Login Response` and close the connection if the login request does not list all CXA matching units. If the flag is set to S (Skip), Cboe Australia will exclude messages from unspecified matching units during replay. If the flag is set to R (Replay), Cboe Australia will send messages from unspecified units during replay (starting with sequence 1 per unspecified unit). Cboe Australia will send a `Replay Complete` message when replay is finished. If there are no messages to replay, a `Replay Complete` message will be sent immediately after a `Login Response` message. Cboe Australia will reject all orders during replay.

If a participant has requested replay messages using a properly formatted `Login Request` after a disconnect, any unacknowledged orders remaining with the participant after the `Replay Complete` message is received should be assumed to be unknown to Cboe Australia.

Unsequenced messages will not be included during replay.

A session is identified by the *SessionId* and *SessionSubId* (both supplied by Cboe Australia).

If a login is rejected an appropriate `Login Response` message will be sent and the connection will be terminated.

See 'Appendix A – Login Strategies' for recommended login messaging.

## 2.3 Sequence Reset

A reset sequence operation is not available for Binary Order Entry. However, a participant can send a `Login Request` message with *ReplayUnspecifiedUnit* field set to S (Skip), and *NumberOfUnits* field set to zero. Then, upon receiving a `Login Response` message from Cboe Australia, the participant can use the *ClientSequence* value in that message as the sequence starting point for sending future messages.

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

## 2.4 Heartbeats

`Client Heartbeat` messages are sent from participant to Cboe Australia, and `Server Heartbeat` messages are sent from Cboe Australia to Participant if no other data has been sent in that direction for one second. Similar to other session level messages, heartbeats from Cboe Australia to the participant do not increment the sequence number. If Cboe Australia receives no inbound data or heartbeats for five seconds, a `Logout Response` message will be sent, and the connection will be terminated. **Clients are encouraged to have a one second heartbeat interval and to perform similar connection staleness logic.**

## 2.5 Logging Out

To gracefully log out of a session, a `Logout Request` message should be sent by the participant. Cboe Australia will finish sending any queued data for that port, respond with a `Logout Response` message, and close the connection. After receipt of a `Logout Request` message, Cboe Australia will ignore all other inbound (Participant to Cboe Australia) messages except for `Client Heartbeat`.

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

# 3. Session Messages

**Table 1 - Session Message Types**

| Direction | Message Name | Type | Sequenced |
|---|---|---|---|
| Participant to Cboe | Login Request | 01 00 (1) | No |
| Participant to Cboe | Logout Request | 02 00 (2) | No |
| Participant to Cboe | Client Heartbeat | 03 00 (3) | No |
| Cboe to Participant | Login Response | F5 01 (501) | No |
| Cboe to Participant | Replay Complete | F6 01 (502) | No |
| Cboe to Participant | Logout Response | F7 01 (503) | No |
| Cboe to Participant | Server Heartbeat | F8 01 (504) | No |

## 3.1 Participant to Cboe Australia

### 3.1.1 Login Request

A `Login Request` message must be sent as the first message upon connection.

This message includes a repeating group starting with field *UnitNumber* which repeats *NumberOfUnits* times. *NumberOfUnits* can be used to specify the last consumed sequence numbers per matching unit received by the participant. Cboe Australia uses these sequence numbers to determine what outbound (Cboe Australia to participant) traffic, if any, was missed by the participant. If *NumberOfUnits* is 0, it is assumed the participant has not received any messages (e.g., start of day).

The participant does not need to include a sequence number for a unit if they never received messages from it. If the participant wishes to send a value for the unit anyway, 0 is the only allowed value.

| Field Name | Offset | Length | Data Type | Description |
|---|---|---|---|---|
| StartOfMessage | 0 | 2 | Binary | B0 E3 (58288) |
| MessageLength | 2 | 2 | Binary | (30 + 5*NumberOfUnits) |
| MessageType | 4 | 2 | Binary | 01 00 (1) |
| MatchingUnit | 6 | 1 | Binary | Must be zero |
| Reserved | 7 | 1 | Binary | Must be zero |
| SequenceNumber | 8 | 4 | Binary | Must be zero |
| SessionId | 12 | 4 | AlphaNumeric | Session Id as supplied by Cboe Australia |
| SessionSubId | 16 | 4 | AlphaNumeric | Session Sub Id as supplied by Cboe Australia (ex. “0001”) |
| Password | 20 | 10 | AlphaNumeric | The password associated with the *SessionId* and *SessionSubId*. |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| ReplayUnspecifiedUnit | 30 | 1 | Text | Controls replay behavior for unknown units. Must be one of:<br>F = fail if any unit is not specified<br>R = replay any unspecified unit from one<br>S = skip replay of unspecified units |
| NumberOfUnits | 31 | 1 | Binary | The number (possibly 0) of unit/sequence pairs to follow, one per unit from which the participant has received messages over this port. This value must be 0, 1, or 2 since Cboe Australia has 2 matching units. |
| →UnitNumber | 32 | 1 | Binary | A unit number (1 or 2, since Cboe Australia has 2 matching units) |
| →UnitSequence | 33 | 4 | Binary | Last received sequence number for the unit (Zero if nothing has been received). |

#### *3.1.1.1 Example Login Request Message*

| | | |
|---|---|---|
| StartOfMessage | BO E3 | |
| MessageLength | 1E 00 | 30 |
| MessageType | 01 00 | 1 (login request) |
| MatchingUnit | 00 | |
| Reserved | 00 | |
| SequenceNumber | 00 | (unsequenced) |
| SessionId | 41 42 43 44 | (ex. “ABCD”) - Session Id as supplied by Cboe Australia |
| SessionSubId | 30 30 30 31 | (ex. “0001”) Session Sub Id as supplied by Cboe Australia |
| Password | 61 62 63 64 65 66 67 68 69 6A | (ex. “abcdefghij”) Password |
| ReplayUnspecifiedUnit | 52 | “R” - Replay unknown units. |

### 3.1.2 Logout Request

To end the session, the participant should send a `Logout Request` message. Cboe Australia will finish sending any queued data, respond with a `Logout Response` message, and close the connection.

A participant may simply close the connection without logging out however, queued messages may be lost when closing the connection without logging out.

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| Field Name | Offset | Length | Data Type | Description |
|---|---|---|---|---|
| StartOfMessage | 0 | 2 | Binary | B0 E3 (58288) |
| MessageLength | 2 | 2 | Binary | 0A 00 (10) |
| MessageType | 4 | 2 | Binary | 02 00 (2) |
| MatchingUnit | 6 | 1 | Binary | Must be zero |
| Reserved | 7 | 1 | Binary | Must be zero |
| SequenceNumber | 8 | 4 | Binary | Must be zero |

### 3.1.3 Example Logout Request Message

| | | |
|---|---|---|
| StartOfMessage | B0 E3 | Start of message bytes |
| MessageLength | 0A 00 | 10 bytes |
| MessageType | 02 00 | Logout Request |
| MatchingUnit | 00 | Must be zero |
| Reserved | 00 | Must be zero |
| SequenceNumber | 00 00 00 00 | Must be zero |

### 3.1.4 Client Heartbeat

See 'Section 2.4 - Heartbeats' for more information about heartbeats and the session level protocol.

| Field Name | Offset | Length | Data Type | Description |
|---|---|---|---|---|
| StartOfMessage | 0 | 2 | Binary | B0 E3 (58288) |
| MessageLength | 2 | 2 | Binary | 0A 00 (10) |
| MessageType | 4 | 2 | Binary | 03 00 (3) |
| MatchingUnit | 6 | 1 | Binary | Must be zero |
| Reserved | 7 | 1 | Binary | Must be zero |
| SequenceNumber | 8 | 4 | Binary | Must be zero |

#### *3.1.4.1 Example Client Heartbeat Message*

| | | |
|---|---|---|
| StartOfMessage | B0 E3 | Start of message bytes |
| MessageLength | 0A 00 | 10 bytes |
| MessageType | 03 00 | Client Heartbeat |
| MatchingUnit | 00 | Must be zero |
| Reserved | 00 | Must be zero |
| SequenceNumber | 00 00 00 00 | Must be zero |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

## 3.2 Cboe Australia to Participant

### 3.2.1 Login Response

`A Login Response` message is sent in response to a `Login Request` message. On a successful login, the *LoginResponseStatus* will be set to 'A'. On a failed login, *LoginResponseStatus* will be set to a value other than 'A', and *LoginResponseText* will be set to an appropriate failure description.

Note that the repeating group starting with field *UnitNumber* provides the highest available Cboe Australia to participant sequence number for the unit. All units will be included in a successful `Login Response`, regardless of whether all units were listed in the `Login Request` message.

| Field Name | Offset | Length | Data Type | Description |
|---|---|---|---|---|
| StartOfMessage | 0 | 2 | Binary | `B0 E3` (58288) |
| MessageLength | 2 | 2 | Binary | (76 + NumberOfUnits*5) |
| MessageType | 4 | 2 | Binary | `F5 01` (501) |
| MatchingUnit | 6 | 1 | Binary | Will be zero |
| Reserved | 7 | 1 | Binary | Unspecified |
| SequenceNumber | 8 | 4 | Binary | Will be zero |
| LoginResponseStatus | 12 | 1 | Text | Accepted, or the reason for the rejection.<br>`A` = Accepted<br>`B` = Session in Use<br>`D` = Disabled<br>`I` = Invalid Unit<br>`M` = Invalid Message (first message was not Login Request)<br>`N` = Not Authorised<br>`Q` = Sequence Ahead<br>`S` = Invalid Session |
| LoginResponseText | 13 | 60 | Text | Human-readable text with additional information about the reason for rejection. ASCII NUL (0x00) filled on the right, if necessary. |
| ClientSequence | 73 | 4 | Binary | Last inbound (participant to Cboe Australia) message sequence number processed by Cboe Australia on this port. |
| NumberOfUnits | 77 | 1 | Binary | A number, n, of unit/sequence pairs to follow, one per unit. A pair for every unit will be sent, even if no messages have been sent to this port today. For unsuccessful logins, this will be 0. |
| →UnitNumber | 78 | 1 | Binary | A unit number |
| →UnitSequence | 79 | 4 | Binary | Highest available Cboe Australia to participant sequence number for the unit. |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

#### *3.2.1.1 Example Login Response Message*

| | | |
|---|---|---|
| StartOfMessage | B0 E3 | Start of message bytes |
| MessageLength | 51 00 | 81 bytes |
| MessageType | F5 01 | Login Response |
| MatchingUnit | 00 | Always zero |
| Reserved | 00 | |
| SequenceNumber | 00 00 00 00 | Always zero |
| LoginResponseStatus | 41 | "A" (Accepted) |
| LoginResponseText | 54 45 53 54 49 4E 47 00<br>00 00 00 00 00 00 00 00<br>00 00 00 00 00 00 00 00<br>00 00 00 00 00 00 00 00<br>00 00 00 00 00 00 00 00<br>00 00 00 00 00 00 00 00<br>00 00 00 00 00 00 00 00<br>00 00 00 00 | "TESTING" |
| ClientSequence | 01 00 00 00 | 1 |
| NumberOfUnits | 01 | 1 unit |
| →UnitNumber | 01 | Unit 1 |
| →UnitSequence | 06 41 27 00 | 160,790 |
| →UnitNumber | 02 | Unit 2 |
| →UnitSequence | 3F 15 00 00 | 5,439 |

### 3.2.2 Replay Complete

See 'Section 2.2 - Login, Replay, and Sequencing' for more information about replay and the session level protocol.

| Field Name | Offset | Length | Data Type | Description |
|---|---|---|---|---|
| StartOfMessage | 0 | 2 | Binary | B0 E3 (58288) |
| MessageLength | 2 | 2 | Binary | 0A (10) |
| MessageType | 4 | 2 | Binary | F6 01 (502) |
| MatchingUnit | 6 | 1 | Binary | Will be zero |
| Reserved | 7 | 1 | Binary | Unspecified |
| SequenceNumber | 8 | 4 | Binary | Will be zero |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

#### *3.2.2.1 Example Replay Complete Message*

| | | |
|---|---|---|
| StartOfMessage | B0 E3 | Start of message bytes |
| MessageLength | 0A 00 | 10 bytes |
| MessageType | F6 01 | Replay Complete |
| MatchingUnit | 00 | Always zero |
| Reserved | 00 | |
| SequenceNumber | 00 00 00 00 | Always zero |

### 3.2.3 Logout Response

A `Logout Response` is usually sent in response to a `Logout Request`. Any queued data is transmitted, a `Logout Response` is sent, and Cboe Australia will close the connection. However, a `Logout Response` may also be sent if the participant violates the protocol specification (e.g., by moving backwards in sequence number).

| Field Name | Offset | Length | Data Type | Description |
|---|---|---|---|---|
| StartOfMessage | 0 | 2 | Binary | `B0 E3` (58288) |
| MessageLength | 2 | 2 | Binary | `47 00` (71 ) |
| MessageType | 4 | 2 | Binary | `F7 01` (503) |
| MatchingUnit | 6 | 1 | Binary | Will be zero |
| Reserved | 7 | 1 | Binary | Unspecified |
| SequenceNumber | 8 | 4 | Binary | Will be zero |
| LogoutReason | 12 | 1 | Text | Reason for the logout.<br>U = User Requested<br>A = Administrative<br>! = Protocol Violation (ex. timeout due to inactivity) |
| LogoutReasonText | 13 | 60 | Text | Human-readable text with additional information about the reason for logout. |

#### *3.2.3.1 Example Logout Response Message*

| | | |
|---|---|---|
| StartOfMessage | B0 E3 | Start of message bytes |
| MessageLength | 47 00 | 71 bytes |
| MessageType | F7 01 | Logout Response |
| MatchingUnit | 00 | Always zero |
| Reserved | 00 | |
| SequenceNumber | 00 00 00 00 | Always zero |
| LogoutReason | 55 | "`U`" (User Requested) |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | |
|---|---|---|
| LogoutReasonText | 54 45 53 54 49 4E 47 00<br>00 00 00 00 00 00 00 00<br>00 00 00 00 00 00 00 00<br>00 00 00 00 00 00 00 00<br>00 00 00 00 00 00 00 00<br>00 00 00 00 00 00 00 00<br>00 00 00 00 00 00 00 00<br>00 00 00 00 | “TESTING” |

### 3.2.4 Server Heartbeat

See ‘Section 2.4 - Heartbeats’ for more information about heartbeats and the session level protocol.

| Field Name | Offset | Length | Data Type | Description |
|---|---|---|---|---|
| StartOfMessage | 0 | 2 | Binary | `B0 E3` (58288) |
| MessageLength | 2 | 2 | Binary | `0A` (10) |
| MessageType | 4 | 2 | Binary | `F8 01` (504) |
| MatchingUnit | 6 | 1 | Binary | Will be zero |
| Reserved | 7 | 1 | Binary | Unspecified |
| SequenceNumber | 8 | 4 | Binary | Will be zero |

#### *3.2.4.1 Example Server Heartbeat Message*

| | | |
|---|---|---|
| StartOfMessage | B0 E3 | Start of message bytes |
| MessageLength | 0A 00 | 10 bytes |
| MessageType | F8 01 | Server Heartbeat |
| MatchingUnit | 00 | Always zero |
| Reserved | 00 | |
| SequenceNumber | 00 00 00 00 | Always zero |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

# 4. Application Messages

Note that if Cboe Australia receives a message type not listed in this specification, the connection will be closed. See section "2.1.1 - Handling of Invalid Message Headers" for details.

**Table 2 - Application Message Types**

| Direction | Message Name | Version | Type | Seq. |
|---|---|---|---|---|
| to Cboe | NewOrderAUEquitiesV1 | V1 | 3001 | Y |
| to Cboe | ModifyOrderAUEquitiesV1 | V1 | 3002 | Y |
| to Cboe | CancelOrderAUEquitiesV1 | V1 | 3003 | Y |
| to Cboe | PurgeOrdersAUEquitiesV1 | V1 | 3004 | Y |
| to Cboe | NewOrderAUEquitiesV1 | V2 | 3005 | Y |
| to Cboe | ModifyOrderAUEquitiesV2 | V2 | 3006 | Y |
| from Cboe | OrderAcknowledgementAUEquitiesV1 | V1 | 3501 | Y |
| from Cboe | OrderRejectedAUEquitiesV1 | V1 | 3502 | N |
| from Cboe | OrderModifiedAUEquitiesV1 | V1 | 3503 | Y |
| from Cboe | ModifyRejectedAUEquitiesV1 | V1 | 3504 | N |
| from Cboe | OrderRestatedAUEquitiesV1 | V1 | 3505 | Y |
| from Cboe | OrderExecutionAUEquitiesV1 | V1 | 3506 | Y |
| from Cboe | OrderCancelledAUEquitiesV1 | V1 | 3507 | Y |
| from Cboe | CancelRejectedAUEquitiesV1 | V1 | 3508 | N |
| from Cboe | ExecutionRestatementAUEquitiesV1 | V1 | 3509 | Y |
| from Cboe | PurgeAcknowledgementAUEquitiesV1 | V1 | 3510 | N |
| from Cboe | PurgeRejectedAUEquitiesV1 | V1 | 3511 | N |
| from Cboe | TradeCancelCorrectAUEquitiesV1 | V1 | 3512 | Y |
| from Cboe | OrderModifiedAUEquitiesV2 | V2 | 3513 | Y |
| from Cboe | CarriedRestatementAUEquitiesV1 | V1 | 3514 | Y |
| from Cboe | DoneForDayAUEquitiesV1 | V1 | 3515 | Y |

## 4.1 Participant to Cboe Australia

### 4.1.1 New Order

The Trading Participant Identifier for the order will be assigned via the port configuration.

#### *4.1.1.1 NewOrderAUEquitiesV1*

| Field Name | Offset | Length | Data Type | Description |
|---|---|---|---|---|
| StartOfMessage | 0 | 2 | Binary | B0 E3 (58288) |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| MessageLength | 2 | 2 | Binary | 170 |
| MessageType | 42 | 2 | Binary | B9 0B (3001) |
| MatchingUnit | 62 | 1 | Binary | Zero - from participant message |
| Reserved | 7 | 1 | Binary | MUST be zero from participant |
| SequenceNumber | 8 | 4 | Binary | *(Sequenced by Participant)* |
| Account | 12 | 10 | Text | Used by a trading participant to provide a broker cross reference.<br>Valid characters are ASCII 32-126.<br>Reflected back on messages associated with this order. |
| TradingParticipantID | 22 | 2 | Binary | The (4 digit) trading participant identifier.<br>A zero value will be populated with the port default (see Port Attributes section) |
| ClOrdID | 24 | 20 | Text | Unique Id chosen by the participant. Characters in the ASCII range 33-126 are allowed, except for **comma, semicolon, pipe, 'at' (@)**, and **double quotes (").** If the ClOrdID matches a live order, the order will be rejected as duplicate. A leading **tilde** (~) cannot be sent on any *ClOrdID* and will result in a reject. These are reserved for internal use by Cboe Australia and could be received because of a Cboe-generated *ClOrdID*.<br>**Note:** Although Cboe Australia only enforces uniqueness of *ClOrdID* values among currently live orders, using unique *ClOrdID* values is strongly recommended. |
| ExecInst | 44 | 1 | Text | All values not defined below are rejected.<br>Ignored if blank (0x00).<br>P = Market Peg (peg Buy [Sell] to NBBO Offer [Bid], one tick within NBBO)<br>R = Primary Peg (peg Buy [Sell] to NBBO Bid [Offer], one tick within NBBO)<br>M = Midpoint (peg to NBBO Midpoint)<br>j = BestX Order (insert the limit order at the best visible price on same side). Not applicable to IOC or FOK.<br>k = MarketX Order (insert the limit order against the best visible price on opposite side). |
| OrderQty | 45 | 4 | Binary | Order quantity. System limit is 999,999,999. |
| OrdType | 49 | 1 | Text | 2 = Limit |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| | | | | 5 = Market on Close (MOC)<br>P = Pegged<br>MOC requires *MaxFloor* to be 0 if *TimeInForce* is 0 or 6.<br>Pegged requires *ExecInst* to be P, R, or M, and requires *MaxFloor* to be 0. |
| Price | 50 | 8 | BinaryPrice | If *OrdType* is 2 (limit), *Price* must be an applicable minimum trading increment.<br>If *OrdType* is P (peg), *Price* must specify a cap price as an applicable minimum trading increment, or a ½ increment if *ExecInst*=M (mid).<br>If OrdType is 5 (MOC), Price must be 0. |
| Capacity | 58 | 1 | Text | The capacity for the order. Required on all orders.<br>A = Agency<br>P = Principal<br>X = Mixed |
| Side | 59 | 1 | Text | 1 = Buy<br>2 = Sell<br>5 = Sell Short |
| Symbol | 60 | 6 | Alphanumeric | Instrument symbol (upper case alphanumeric) |
| TimeInForce | 66 | 1 | Text | 0 = DAY (default)<br>1 = GTC (Good Till Cancel).<br>3 = IOC (portion not filled immediately is cancelled).<br>4 = FOK (an IOC where the entire size must be filled immediately, otherwise the order will be cancelled back).<br>6 = GTD. Expires at the specified: *ExpireTime* or end of day, or *ExpireDate* for multi-day orders. |
| MinQty | 67 | 4 | Binary | Minimum fill quantity for orders. Required if *MEQSE* is true. Accepted only on IOC, MOC, and Pegged orders.<br>If MEQSE is false, multiple consecutive fills may be created such that their combined size satisfies the *MinQty* requirement.<br>When the remaining size on an order is less than the defined *MinQty*, LeavesQty *is* treated as the MinQty. |
| MaxFloor | 71 | 4 | Binary | Portion of *OrderQty* to display. Used for iceberg orders and hidden orders. |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| | | | | Limit orders (*OrdType* = 2) having *MaxFloor* > 0 are iceberg orders. For iceberg orders, *MaxFloor* is the displayed quantity. The balance is reserve. The displayed quantity of each order at a price level is decremented first.<br><br>Limit orders (*OrdType*=2) having *MaxFloor* = 0 will display their full size.<br><br>Pegged orders (*OrdType* = P) and MOC orders (*OrdType* = 5) must use *MaxFloor* =0, which hides the entire quantity.<br><br>**Ignored** if order is IOC or FOK. |
| ExpireTime | 75 | 8 | DateTime | Required for GTD order (*TimeInForce* = 6), *ExpireTime* specifies the date-time (in UTC) when the order expires. GTD orders will expire at the earlier of the end of the current trading day and the *ExpireTime*. |
| ClearingParticipantID | 83 | 4 | Binary | The CHESS clearing participant identifier. Must be a valid 5-digit CHESS PID. |
| ClearingAccount | 87 | 10 | Alphanumeric | Supplemental identifier.<br>Recorded and made available on order acknowledgements. Available via Drop feeds.<br><br>This field may be blank.<br><br>10 character alphanumeric. Spaces, symbols, or special characters are not allowed.<br>This field is not sent to CHESS. |
| Undisclosed | 97 | 1 | Boolean | If true, indicates an undisclosed order.<br>Y = True<br>N = False |
| ClientCrossRef | 98 | 15 | Text | Can be used to provide additional participant-side cross reference information. |
| ShortSellNakedQty | 113 | 4 | Binary | Defines the portion of OrderQty (38) that is naked.<br><br>Ignored unless *Side* = 5 |
| ShortSellCoveredQty | 117 | 4 | Binary | Defines the portion of *OrderQty* (38) that is covered.<br><br>Ignored unless *Side* = 5 |
| ShortSellLongQty | 121 | 4 | Binary | Defines the portion of *OrderQty* (38) that is owned by the participant who placed the order. |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| | | | | Orders having *Side* = 5 (sell short) will be rejected unless *ShortSellNakedQty* + *ShortSellCoveredQty* + *ShortSellLongQty* is equal to the *OrderQty* and ShortSellLongQty < *OrderQty*.<br><br>Ignored unless *Side* = 5 |
| RiskGroupID | 125 | 2 | Binary | Used to group orders for use in order purges where multiple orders can be cancelled by specifying a list of *RiskGroupIDs*.<br><br>A zero value is treated as "no *RiskGroupID*". |
| NoTradeFeat | 127 | 1 | Text | Ignored if *NoTradeKey* is not set. Declares the handling of self-trading.<br>`U` = No self trade protection<br>`N` = Cancel Newest<br>`O` = Cancel Oldest<br>`B` = Cancel Both<br>`S` = Cancel Smallest<br>`D` = Decrement larger/Cancel Smaller<br>`d` = Same as D above, but only decrement *LeavesQty*. Do not restate *OrderQty*.<br>`X` = Booking Purpose Trade; the resultant trade is reported to CHESS as a booking purpose trade and as a cancel on public market data feeds.<br>`NULL (0x00)` = Apply the port default.<br><br>If the older and newer orders in a self-trade have conflicting instructions, the instruction of the newer order will be honoured, except that if the inbound order specifies Decrement and the resting order does not, and the resting order is larger, then both orders will be cancelled. This exception is to protect the order entry software for the resting order from receiving an unexpected restatement message. |
| NoTradeKey | 128 | 10 | Text | Interactions of orders from the same participant (defined by reference to the first 3 digits of the trading PID) with the same *NoTradeKey* value will be managed by the newer order's *NoTradeFeat* instruction. |
| MEQSE | 138 | 1 | Text | If true, then any single fill will be for at least *MinQty* shares.<br>`Y` = True<br>`N` = False |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| | | | | If True, then the order will be rejected unless *MinQty* is also specified or the *OrdType* is not 5 or P (MOC or Pegged). |
| OriginOfTrans | 139 | 20 | Text | Market Integrity Rule defined field for Origin of Order (ex. ACN, ABN, CHESS HIN). |
| IntermediaryID | 159 | 10 | Text | Market Integrity Rule defined field for Intermediary Identifier–may be populated with AFS license number.<br>Characters in ASCII range 32-126 are allowed. |
| DirectedWholesaleIndicator | 169 | 1 | Boolean | Market Integrity Rule defined field to identify wholesale participant.<br>Y = True<br>N = False |
| RemoveCrossingFromClearing | 170 | 1 | Boolean | If "Y", instructs Cboe Australia on how to process crossed trades with CHESS. Please contact Cboe Australia to confirm settings.<br>Y = True<br>N = False |
| BrokerPreferencing | 171 | 1 | Text | Y = match with other orders using broker preferencing when possible.<br>N = do not apply broker preferencing<br>NULL (0x00) = apply port default |

#### *4.1.1.2 NewOrderAUEquitiesV2*

| Field Name | Offset | Length | Data Type | Description |
|---|---|---|---|---|
| StartOfMessage | 0 | 2 | Binary | B0 E3 (58288) |
| MessageLength | 2 | 2 | Binary | 176 |
| MessageType | 4 | 2 | Binary | BD 0B (3005) |
| MatchingUnit | 6 | 1 | Binary | Zero - from participant message |
| Reserved | 7 | 1 | Binary | MUST be zero from participant |
| SequenceNumber | 8 | 4 | Binary | *(Sequenced by Participant)* |
| Account | 12 | 10 | Text | Used by a trading participant to provide a broker cross reference.<br>Valid characters are ASCII 32-126.<br>Reflected back on messages associated with this order. |
| TradingParticipantID | 22 | 2 | Binary | The (4 digit) trading participant identifier.<br>A zero value will be populated with the port default (see Port Attributes section) |
| ClOrdID | 24 | 20 | Text | Unique Id chosen by the participant.<br>Characters in the ASCII range 33-126 are |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| | | | | allowed, except for **comma, semicolon, pipe, 'at' (@)**, and **double quotes (").** If the ClOrdID matches a live order, the order will be rejected as duplicate. A leading **tilde** (~) cannot be sent on any *ClOrdID* and will result in a reject. These are reserved for internal use by Cboe Australia and could be received because of a Cboe-generated *ClOrdID*.<br><br>**Note:** Although Cboe Australia only enforces uniqueness of *ClOrdID* values among currently live orders, using unique *ClOrdID* values is strongly recommended. |
| ExecInst | 44 | 1 | Text | All values not defined below are rejected.<br>Ignored if blank (0x00).<br>P = Market Peg (peg Buy [Sell] to NBBO Offer [Bid], one tick within NBBO)<br>R = Primary Peg (peg Buy [Sell] to NBBO Bid [Offer], one tick within NBBO)<br>M = Midpoint (peg to NBBO Midpoint)<br>j = BestX Order (insert the limit order at the best visible price on same side). Not applicable to IOC or FOK.<br>k = MarketX Order (insert the limit order against the best visible price on opposite side). |
| OrderQty | 45 | 4 | Binary | Order quantity. System limit is 999,999,999. |
| OrdType | 49 | 1 | Text | 2 = Limit<br>5 = Market on Close (MOC)<br>P = Pegged<br><br>MOC requires *MaxFloor* to be 0 if *TimeInForce* is 0 or 6.<br><br>Pegged requires *ExecInst* to be P, R, or M, and requires *MaxFloor* to be 0. |
| Price | 50 | 8 | BinaryPrice | If *OrdType* is 2 (limit), *Price* must be an applicable minimum trading increment.<br>If *OrdType* is P (peg), *Price* must specify a cap price as an applicable minimum trading increment, or a ½ increment if *ExecInst*=M (mid).<br><br>If OrdType is 5 (MOC), Price must be 0. |
| Capacity | 58 | 1 | Text | The capacity for the order. Required on all orders.<br>A = Agency<br>P = Principal |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| | | | | X = Mixed |
| Side | 59 | 1 | Text | 1 = Buy<br>2 = Sell<br>5 = Sell Short |
| Symbol | 60 | 6 | Alphanumeric | Instrument symbol (upper case alphanumeric) |
| TimeInForce | 66 | 1 | Text | 0 = DAY (default).<br>1 = GTC (Good Till Cancel).<br>3 = IOC (portion not filled immediately is cancelled).<br>4 = FOK (an IOC where the entire size must be filled immediately, otherwise the order will be cancelled back).<br>6 = GTD. Expires at the specified:<br>• *ExpireTime* or end of day, or<br>• *ExpireDate* for multi-day orders. |
| MinQty | 67 | 4 | Binary | Minimum fill quantity for orders. Required if *MEQSE* is true. Accepted only on IOC, MOC, and Pegged orders.<br>If MEQSE is false, multiple consecutive fills may be created such that their combined size satisfies the *MinQty* requirement.<br>When the remaining size on an order is less than the defined *MinQty*, LeavesQty *is* treated as the MinQty. |
| MaxFloor | 71 | 4 | Binary | Portion of *OrderQty* to display. Used for iceberg orders and hidden orders.<br>Limit orders (*OrdType* = 2) having *MaxFloor* > 0 are iceberg orders. For iceberg orders, *MaxFloor* is the displayed quantity. The balance is reserve. The displayed quantity of each order at a price level is decremented first.<br>Limit orders (*OrdType*=2) having *MaxFloor* = 0 will display their full size.<br>Pegged orders (*OrdType* = P) and MOC orders (*OrdType* = 5) must use *MaxFloor* =0, which hides the entire quantity.<br>**Ignored** if order is IOC or FOK. |
| ExpireTime | 75 | 8 | DateTime | Required for GTD *order* (TimeInForce = 6), ExpireTime specifies the date-time (in UTC) when the order expires. GTD orders will expire at the earlier of the end of the current trading day and the ExpireTime. |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| ClearingParticipantID | 83 | 4 | Binary | The CHESS clearing participant identifier. Must be a valid 5-digit CHESS PID. |
| ClearingAccount | 87 | 10 | Alphanumeric | Supplemental identifier.<br>Recorded and made available on order acknowledgements. Available via Drop feeds.<br>This field may be blank.<br>10 character alphanumeric. Spaces, symbols, or special characters are not allowed.<br>This field is not sent to CHESS. |
| Undisclosed | 97 | 1 | Boolean | If true, indicates an undisclosed order.<br>Y = True<br>N = False |
| ClientCrossRef | 98 | 15 | Text | Can be used to provide additional participant-side cross reference information. |
| ShortSellNakedQty | 113 | 4 | Binary | Defines the portion of OrderQty (38) that is naked.<br>Ignored unless *Side* = 5 |
| ShortSellCoveredQty | 117 | 4 | Binary | Defines the portion of *OrderQty* (38) that is covered.<br>Ignored unless *Side* = 5 |
| ShortSellLongQty | 121 | 4 | Binary | Defines the portion of *OrderQty* (38) that is owned by the participant who placed the order.<br>Orders having *Side* = 5 (sell short) will be rejected unless *ShortSellNakedQty* + *ShortSellCoveredQty* + *ShortSellLongQty* is equal to the *OrderQty* and ShortSellLongQty < *OrderQty*.<br>Ignored unless *Side* = 5 |
| RiskGroupID | 125 | 2 | Binary | Used to group orders for use in order purges where multiple orders can be cancelled by specifying a list of *RiskGroupIDs*.<br>A zero value is treated as “no *RiskGroupID*”. |
| NoTradeFeat | 127 | 1 | Text | Ignored if *NoTradeKey* is not set. Declares the handling of self-trading.<br>U = No self trade protection<br>N = Cancel Newest<br>O = Cancel Oldest<br>B = Cancel Both<br>S = Cancel Smallest |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| | | | | D = Decrement larger/Cancel Smaller<br>d = Same as D above, but only decrement *LeavesQty*. Do not restate *OrderQty*.<br>X = Booking Purpose Trade; the resultant trade is reported to CHESS as a booking purpose trade and as a cancel on public market data feeds.<br>NULL (0x00) = Apply the port default.<br><br>If the older and newer orders in a self-trade have conflicting instructions, the instruction of the newer order will be honoured, except that if the inbound order specifies Decrement and the resting order does not, and the resting order is larger, then both orders will be cancelled. This exception is to protect the order entry software for the resting order from receiving an unexpected restatement message. |
| NoTradeKey | 128 | 10 | Text | Interactions of orders from the same participant (defined by reference to the first 3 digits of the trading PID) with the same *NoTradeKey* value will be managed by the newer order's *NoTradeFeat* instruction. |
| MEQSE | 138 | 1 | Text | If true, then any single fill will be for at least *MinQty* shares.<br>Y = True<br>N = False<br>If True, then the order will be rejected unless *MinQty* is also specified or the *OrdType* is not 5 or P (MOC or Pegged). |
| OriginOfTrans | 139 | 20 | Text | Market Integrity Rule defined field for Origin of Order (ex. ACN, ABN, CHESS HIN). |
| IntermediaryID | 159 | 10 | Text | Market Integrity Rule defined field for Intermediary Identifier–may be populated with AFS license number.<br>Characters in ASCII range 32-126 are allowed. |
| DirectedWholesaleIndicator | 169 | 1 | Boolean | Market Integrity Rule defined field to identify wholesale participant.<br>Y = True<br>N = False |
| RemoveCrossingFromClearing | 170 | 1 | Boolean | If "Y", instructs Cboe Australia on how to process crossed trades with CHESS. Please contact Cboe Australia to confirm settings.<br>Y = True |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| | | | | N = False |
| BrokerPreferencing | 171 | 1 | Text | Y = match with other orders using broker preferencing when possible.<br>N = do not apply broker preferencing<br>NULL (0x00) = apply port default |
| RoutingInst | 172 | 1 | Text | P = Post Only (never take liquidity) where *OrdType*=2 (Limit Order) and *TimeInForce*=0 (DAY) or 6 (GTD).<br>F = Focused Nearpoint (only eligible to trade with opposing Farpoint) where *OrdType*=P (Pegged order) and *ExecInst*=R (Primary / Nearpoint).<br>Defaults to port setting if NULL (0x00) |
| DisplayIndicator | 173 | 1 | Text | Treatment of Post Only orders on entry if creating a locked NBBO. All Post Only orders on entry which would create a crossed NBBO are cancelled back.<br>V = Post Only orders which would lock the NBBO on entry are allowed to do so.<br>R = Cancel back the Post Only order on entry if it were to create a locked NBBO.<br>Defaults to port setting if NULL (0x00). |
| ExpireDate | 174 | 4 | Date | Only applicable for *TimeInForce* (59) = '6' (GTD) orders, specifies the date in "YYYYMMDD" format when the order expires. GTD orders with *ExpireDate* will expire at the earlier of the GTC Date Refresh and the *ExpireDate*.<br>Required if *TimeInForce* is 6 (GTD) and *ExpireTime* is not specified.<br>Ignored if *TimeInForce* is not 6 (GTD) or if *ExpireTime* is present. |

### 4.1.2 Modify Order

Request to modify an order. Only *Price*, *OrderQty*, *MinQty*¸ *MaxFloor and ExpireDate* may be adjusted. Modifies will result in a loss of time priority unless the modification involves only one or more of the following:

- a decrease in *OrderQty*,

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

- a change to *MaxFloor* (for Iceberg orders),
- a change to *ExpireDate* (GTD orders only)

Changes in *OrderQty* results in an adjustment of the current order's *OrderQty*. The new *OrderQty* does not directly replace the current order's *LeavesQty*. Rather, a delta is computed from the current *OrderQty* and the replacement *OrderQty*. This delta is then applied to the current *LeavesQty*. If the resulting *LeavesQty* is less than or equal to zero, the order is cancelled. This results in safer behavior when the modification request overlaps partial fills for the current order, leaving the Participant in total control of the share exposure of the order.

If *OrderQty* is changed for a Sell Short order (*Side*=5), then the *ShortSellLongQuantity*, *ShortSellNakedQuantity*, and *ShortSellCoveredQuantity* fields are required to be correctly populated based on the new *OrderQty*.

The *MinQty* is always processed by the Exchange. A *MinQty* of 0 will cause any existing *MinQty* on the order to be removed. Participants may specify the current *MinQty* of the order to avoid any unwanted change to *MinQty*. Any change to *MinQty* will cause a loss of time priority.

*MaxFloor* may only be changed for Iceberg orders. A change in *MaxFloor* takes effect on the next iceberg reload. It may not be changed for Hidden order types (Pegged or MOC). If *MaxFloor* is 0, then it is ignored.

*ExecInst* may be specified as BestX (j) or MarketX (k) as a pricing option when modifying any visible limit order, irrespective of the *ExecInst* originally entered on the limit order. An ExecInst of BestX(j) or MarketX (k) will always lose time priority. If *Price* is also specified, then the order's limit will be adjusted to the less aggressive of current BBO and the specified *Price*. If *ExecInst* is ASCII NUL (0x00), it is ignored. Other values of *ExecInst*, if specified, must match the value from the original order or the Modify will be rejected.

Multi-day orders will have their expiration date updated when amended as follows:

- For GTC orders, the system-defined 'GTC Date Refresh' number of days from the current day resets the expiration date.
- For GTD orders, where *ExpireDate* was used in the original order and the *ExpireDate* is being amended, the order will have the new *ExpireDate* applied.
- For GTD orders, where *ExpireDate* was used in the original order and the *ExpireDate* is not specified in the amend request, the order's *ExpireDate* remains unchanged.

A maximum of 1,679,615 `Modify Order` requests may be made to a single order each trading day. Once the 1,679,615$^{th}$ modification is made, then the next user-generated message on the order should be a `Cancel Order` request.

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

#### *4.1.2.1 ModifyOrderAUEquitiesV1*

| Field Name | Offset | Length | Data Type | Description |
|---|---|---|---|---|
| StartOfMessage | 0 | 2 | Binary | B0 E3 (58288) |
| MessageLength | 2 | 2 | Binary | 86 |
| MessageType | 4 | 2 | Binary | BA 0B (3002) |
| MatchingUnit | 6 | 1 | Binary | Zero - from participant message |
| Reserved | 7 | 1 | Binary | MUST be zero from participant |
| SequenceNumber | 8 | 4 | Binary | *(Sequenced by Participant)* |
| TradingParticipantID | 12 | 2 | Binary | The (4 digit) trading participant identifier.<br>A zero value will be populated with the port default (see Port Attributes section) |
| ClOrdID | 14 | 20 | Text | Unique Id chosen by the participant. Characters in the ASCII range 33-126 are allowed, except for **comma, semicolon, pipe, 'at' (@)**, and **double quotes (").**<br>If the ClOrdID matches a live order, the order will be rejected as duplicate. A leading **tilde** (~) cannot be sent on any *ClOrdID* and will result in a reject. These are reserved for internal use by Cboe Australia and could be received because of a Cboe-generated *ClOrdID*.<br>**Note:** Although Cboe Australia only enforces uniqueness of *ClOrdID* values among currently live orders, using unique *ClOrdID* values is strongly recommended. |
| ExecInst | 34 | 1 | Text | All values not defined below are rejected. Ignored if blank (0x00).<br>P = Market Peg (peg Buy [Sell] to NBBO Offer [Bid], one tick within NBBO)<br>R = Primary Peg (peg Buy [Sell] to NBBO Bid [Offer], one tick within NBBO)<br>M = Midpoint (peg to NBBO Midpoint)<br>j = BestX Order (insert the limit order at the best visible price on same side). Not applicable to IOC or FOK.<br>k = MarketX Order (insert the limit order against the best visible price on opposite side).<br>The message will be rejected if *ExecInst* is P, R, or M and does not match the current *ExecInst* of the order. |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| OrderQty | 35 | 4 | Binary | Order quantity. System limit is 999,999,999. |
| OrigClOrdID | 39 | 20 | Text | The *ClOrdID* of the original order.<br>In the case of multiple changes to a single order, this will be the *ClOrdID* of the **most recent** accepted change. |
| Price | 59 | 8 | BinaryPrice | For the original order's *OrdType*,<br>If *OrdType* is 2 (limit), *Price* must be an applicable minimum trading increment.<br>If *OrdType* is P (peg), *Price* must specify a cap price as an applicable minimum trading increment, or a ½ increment if *ExecInst*=M (mid).<br>If *OrdType* is 5 (MOC), *Price* must be 0. |
| MinQty | 67 | 4 | Binary | Minimum fill quantity for orders. |
| MaxFloor | 71 | 4 | Binary | If *OrdType* is 2 (limit), *MaxFloor* can be specified to a value > 0. Otherwise, MaxFloor must be 0. |
| ShortSellNakedQty | 75 | 4 | Binary | Defines the portion of O*rderQty* (38) that is naked.<br>Ignored unless *Side* = 5 |
| ShortSellCoveredQty | 79 | 4 | Binary | Defines the portion of *OrderQty* (38) that is covered.<br>Ignored unless *Side* = 5 |
| ShortSellLongQty | 83 | 4 | Binary | Defines the portion of *OrderQty* (38) that is owned by the participant who placed the order.<br>Orders having *Side* = 5 (sell short) will be rejected unless *ShortSellNakedQty* + *ShortSellCoveredQty* + *ShortSellLongQty* is equal to the *OrderQty* and *ShortSellLongQty* < *OrderQty*.<br>Ignored unless *Side* = 5 |
| CancelOrigOnReject | 87 | 1 | Text | Y = Cancel original order if modification fails.<br>N = Leave original order alone.<br>NULL (0x00) = Apply port default |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

#### *4.1.2.2 ModifyOrderAUEquitiesV2*

| Field Name | Offset | Length | Data Type | Description |
|---|---|---|---|---|
| StartOfMessage | 0 | 2 | Binary | B0 E3 (58288) |
| MessageLength | 2 | 2 | Binary | 90 |
| MessageType | 4 | 2 | Binary | BE 0B (3006) |
| MatchingUnit | 6 | 1 | Binary | Zero - from participant message |
| Reserved | 7 | 1 | Binary | MUST be zero from participant |
| SequenceNumber | 8 | 4 | Binary | *(Sequenced by Participant)* |
| TradingParticipantID | 12 | 2 | Binary | The (4 digit) trading participant identifier.<br>A zero value will be populated with the port default (see Port Attributes section). |
| ClOrdID | 14 | 20 | Text | Unique Id chosen by the participant. Characters in the ASCII range 33-126 are allowed, except for **comma, semicolon, pipe, 'at' (@)**, and **double quotes (").**<br>If the ClOrdID matches a live order, the order will be rejected as duplicate. A leading **tilde** (~) cannot be sent on any *ClOrdID* and will result in a reject. These are reserved for internal use by Cboe Australia and could be received because of a Cboe-generated *ClOrdID*.<br>**Note:** Although Cboe Australia only enforces uniqueness of *ClOrdID* values among currently live orders, using unique *ClOrdID* values is strongly recommended. |
| ExecInst | 34 | 1 | Text | All values not defined below are rejected. Ignored if blank (0x00).<br>P = Market Peg (peg Buy [Sell] to NBBO Offer [Bid], one tick within NBBO)<br>R = Primary Peg (peg Buy [Sell] to NBBO Bid [Offer], one tick within NBBO)<br>M = Midpoint (peg to NBBO Midpoint)<br>j = BestX Order (insert the limit order at the best visible price on same side). Not applicable to IOC or FOK.<br>k = MarketX Order (insert the limit order against the best visible price on opposite side).<br>The message will be rejected if *ExecInst* is P, R, or M and does not match the current *ExecInst* of the order. |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| OrderQty | 35 | 4 | Binary | Order quantity. System limit is 999,999,999. |
| OrigClOrdID | 39 | 20 | Text | The *ClOrdID* of the original order.<br>In the case of multiple changes to a single order, this will be the *ClOrdID* of the **most recent** accepted change. |
| Price | 59 | 8 | BinaryPrice | For the original order's *OrdType*,<br>If *OrdType* is 2 (limit), *Price* must be an applicable minimum trading increment.<br>If *OrdType* is P (peg), *Price* must specify a cap price as an applicable minimum trading increment, or a ½ increment if *ExecInst*=M (mid).<br>If *OrdType* is 5 (MOC), *Price* must be 0. |
| MinQty | 67 | 4 | Binary | Minimum fill quantity for orders. |
| MaxFloor | 71 | 4 | Binary | If *OrdType* is 2 (limit), *MaxFloor* can be specified to a value > 0. Otherwise, MaxFloor must be 0. |
| ShortSellNakedQty | 75 | 4 | Binary | Defines the portion of O*rderQty* (38) that is naked.<br>Ignored unless *Side* = 5 |
| ShortSellCoveredQty | 79 | 4 | Binary | Defines the portion of *OrderQty* (38) that is covered.<br>Ignored unless *Side* = 5 |
| ShortSellLongQty | 83 | 4 | Binary | Defines the portion of *OrderQty* (38) that is owned by the participant who placed the order.<br>Orders having *Side* = 5 (sell short) will be rejected unless *ShortSellNakedQty* + *ShortSellCoveredQty* + *ShortSellLongQty* is equal to the *OrderQty* and *ShortSellLongQty* < *OrderQty*.<br>Ignored unless *Side* = 5 |
| CancelOrigOnReject | 87 | 1 | Text | Y = Cancel original order if modification fails.<br>N = Leave original order alone.<br>NULL (0x00) = Apply port default |
| ExpireDate | 88 | 4 | Date | Only applicable for *TimeInForce* (59) = '6' (GTD) orders and where *ExpireDate* was accepted on the new order, specifies the |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| | | | | date in "YYYYMMDD" format when the order expires. GTD orders with *ExpireDate* will expire at the earlier of the GTC Date Refresh and the *ExpireDate*.<br><br>If not specified, when amending a multi-day GTD order which contains ExpireDate, the order's expiration date will remain unchanged. |

### 4.1.3 Cancel Order

Request to cancel a single order.

#### *4.1.3.1 CancelOrderAUEquitiesV1*

| Field Name | Offset | Length | Data Type | Description |
|---|---|---|---|---|
| StartOfMessage | 0 | 2 | Binary | B0 E3 (58288) |
| MessageLength | 2 | 2 | Binary | 32 |
| MessageType | 4 | 2 | Binary | BB 0B (3003) |
| MatchingUnit | 6 | 1 | Binary | Zero - from participant message |
| Reserved | 7 | 1 | Binary | MUST be zero from participant |
| SequenceNumber | 8 | 4 | Binary | *(Sequenced by Participant)* |
| TradingParticipantID | 12 | 2 | Binary | The (4 digit) trading participant identifier.<br><br>A zero value will be populated with the port default (see "**Error! Reference source not found.**. **Error! Reference source not found.**"). |
| OrigClOrdID | 14 | 20 | Text | The *ClOrdID* of the original order. |

### 4.1.4 Purge Orders

Request to cancel a group of orders across all of the participant's sessions. The `Purge Orders` message will only be accepted dedicated purge ports (see "1.5.4.3 - BOE Purge Ports").

Participants may choose to implement one or more of the following filters:

1. *PID* Filter – optionally cancel based on all four digits of the Trading Participant Identifier (PID). Set using first character of *MassCancelInst* and sending *OnBehalfOfCompId*.
2. *Symbol* Filter – optionally cancel based on symbol. Set by sending a valid symbol in the *Symbol* fields. Cannot be combined with *RiskGroupID* filter.

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

3. *RiskGroupID* Filter – optionally cancel based on *RiskGroupID.* A maximum of 10 *RiskGroupID*s may be included on a single `Purge Request` message. Set by populating *RiskGroupIDCnt to* a non-zero value. Cannot be combined with *Symbol* filter.
4. *TargetMatchingUnit* Filter – *CustomGroupID* or *PID* purges with no *Symbol* filter may be directed to a specific matching unit using the MatchingUnit optional field. If *TargetMatchingUnit* is zero or not specified, then these purge types will be sent to all matching units.

Participants may use the second character of *MassCancelInst* to set the acknowledgement style. If a single `Purge Acknowledgement` is selected, then *MassCancelID* must be sent.

The system limits the rate at which identical `Purge Request` messages can be submitted to the system. Requests are restricted to twenty (20) messages per second per port.

An identical purge message is defined as a message having all the same *RiskGroupID*, *Symbol*, *TradingParticipantID*, and *TargetMatchingUnit* field values as a previously received message.

#### *4.1.4.1 PurgeOrdersAUEquitiesV1*

| Field Name | Offset | Length | Data Type | Description |
|---|---|---|---|---|
| StartOfMessage | 0 | 2 | Binary | B0 E3 (58288) |
| MessageLength | 2 | 2 | Binary | 58 + (*RiskGroupIDCnt* * 2) |
| MessageType | 4 | 2 | Binary | BC 0B (3004) |
| MatchingUnit | 6 | 1 | Binary | Zero - from participant message |
| Reserved | 7 | 1 | Binary | MUST be zero from participant |
| SequenceNumber | 8 | 4 | Binary | *(Sequenced by Participant)* |
| TradingParticipantID | 12 | 2 | Binary | The (4 digit) trading participant identifier.<br>Required for service bureau purge ports. |
| MassCancelInst | 14 | 16 | Text | At least one character must be provided (PID Filter). Contiguous characters must be specified up to total length. Truncated (unspecified) characters will default to values indicated below.<br>***1st Character : PID Filter***<br>A = No filtering by PID is performed.<br>F = All orders that were sent under the PID specified in *TradingParticipantID* will be cancelled.<br>***2nd Character : Acknowledgement Style***<br>M = (default) Individual Execution Reports are sent for each cancelled order.<br>S = Single `Purge Acknowledgement` sent once all cancels have been processed. Single `Purge Acknowledgement` will |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| | | | | contain *MassCancelId* and *CancelledOrderCount*. *MassCancelId* must be specified, or the `Purge Request` will be rejected.<br>`B` = Both individual `Execution Reports` and single `Purge Acknowledgement` will be sent. Also requires *MassCancelId* to be specified or the `Purge Request` will be rejected.<br>**3rd Character : Order Scope**<br>`D` = (default) Purge Day orders<br>`M` = Purge GTC and GTD orders |
| MassCancelID | 30 | 20 | Text | User-defined identifier of the mass cancel or purge request. |
| Symbol | 50 | 6 | Alphanumeric | Instrument symbol (upper case alphanumeric) |
| TargetMatchingUnit | 56 | 1 | Binary | Matching unit to which the purge should be restricted. Zero is considered "all matching engines". |
| RiskGroupIDCnt | 57 | 1 | Binary | Number of repeating *RiskGroupID* included in this message. Value 0 to 10. |
| RiskGroupID | 58 | 2 | Binary | Used to group orders for use in order purges where multiple orders can be cancelled by specifying a list of *RiskGroupID*s. |

## 4.2 Cboe Australia to Participant

### 4.2.1 Order Acknowledgement

`Order Acknowledgement` messages are sent in response to a `New Order` message. The message corresponds to a FIX Execution Report with *ExecType* (150) = 0 (New). `Order Acknowledgement` messages do not indicate how or if the order was executed or cancelled. Note that the `Order Acknowledgement` of an IOC order (*TimeInForce* = 3) will be immediately followed by zero or more `Order Execution` (fill) messages and at most one `Order Cancelled` message (for any remaining size).

#### *4.2.1.1 OrderAcknowledgementAUEquitiesV1*

| Field Name | Offset | Length | Data Type | Description |
|---|---|---|---|---|
| StartOfMessage | 0 | 2 | Binary | B0 E3 (58288) |
| MessageLength | 2 | 2 | Binary | **202** |
| MessageType | 4 | 2 | Binary | AD 0D (3501) |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| MatchingUnit | 6 | 1 | Binary | |
| Reserved | 7 | 1 | Binary | Value sent to participant not specified |
| SequenceNumber | 8 | 4 | Binary | *(Sequenced per MatchingUnit by Cboe Australia)* |
| TransactionTime | 12 | 8 | DateTime | The time the event occurred in the Matching Engine (not the time the message was sent). |
| Account | 20 | 10 | Text | (Copied from order) |
| TradingParticipantID | 30 | 2 | Binary | (Copied from order / port default) |
| ClOrdID | 32 | 20 | Text | (Copied from order) |
| ExecInst | 52 | 1 | Text | (Copied from order) |
| OrderID | 53 | 8 | Binary | Order identifier supplied by Cboe Australia. This identifier corresponds to the identifiers used in Cboe Australia market data products. |
| OrderQty | 61 | 4 | Binary | (Copied from order) |
| OrdType | 65 | 1 | Text | (Copied from order) |
| Price | 66 | 8 | BinaryPrice | (Copied from order) |
| Capacity | 74 | 1 | Text | (Copied from order) |
| Side | 75 | 1 | Text | (Copied from order) |
| Symbol | 76 | 6 | Alphanumeric | (Copied from order) |
| TimeInForce | 82 | 1 | Text | (Copied from order) |
| MinQty | 83 | 4 | Binary | (Copied from order) |
| MaxFloor | 87 | 4 | Binary | (Copied from order) |
| ExpireTime | 91 | 8 | DateTime | (Copied from order) |
| ClearingParticipantID | 99 | 4 | Binary | (Copied from order) |
| ClearingAccount | 103 | 10 | Alphanumeric | (Copied from order) |
| Undisclosed | 113 | 1 | Boolean | (Copied from order) |
| ClientCrossRef | 114 | 15 | Text | (Copied from order) |
| ShortSellNakedQty | 129 | 4 | Binary | (Copied from order) |
| ShortSellCoveredQty | 133 | 4 | Binary | (Copied from order) |
| ShortSellLongQty | 137 | 4 | Binary | (Copied from order) |
| RiskGroupID | 141 | 2 | Binary | (Copied from order) |
| NoTradeFeat | 143 | 1 | Text | (Copied from order / port default) |
| NoTradeKey | 144 | 10 | Text | (Copied from order) |
| MEQSE | 154 | 1 | Text | (Copied from order) |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| OriginOfTrans | 155 | 20 | Text | (Copied from order) |
| IntermediaryID | 175 | 10 | Text | (Copied from order) |
| DirectedWholesaleIndicator | 185 | 1 | Boolean | (Copied from order) |
| RemoveCrossingFromClearing | 186 | 1 | Boolean | (Copied from order) |
| BrokerPreferencing | 187 | 1 | Boolean | (Copied from order) |
| WorkingPrice | 188 | 8 | BinaryPrice | The Price at which an order is effectively working in the book. |
| BaseLiquidityIndicator | 196 | 1 | Alphanumeric | `Indicates whether the trade added or removed liquidity.`<br>`A = Added Liquidity`<br>`R = Removed Liquidity -`<br>`Indicates whether the trade added or removed liquidity.`<br>`A = Added Liquidity`<br>`R = Removed Liquidity` |
| SubLiquidityIndicator | 197 | 1 | Text | Additional information about the liquidity of an order. Cboe Australia may add additional values without notice. Participants must gracefully ignore unknown values.<br>`NULL (0x00)` = No Additional Information<br>`I` = Added Hidden Liquidity that was price improved<br>`J` = Added Liquidity NBBO Joiner<br>`K` = Added Liquidity from the hidden (reserve) portion of an iceberg order<br>`S` = Added Liquidity NBBO Setter |
| RoutingInst | 198 | 1 | Text | (Copied from order) |
| DisplayIndicator | 199 | 1 | Text | (Copied from order) |
| ExpireDate | 200 | 4 | Date | Date of order expiration (last day the order can trade) if *TimeInForce* is:<br>• 1 (GTC) or<br>• 6 (GTD) and *ExpireDate* specified. |

*(New fields may be introduced at the end of this message. Consequently, Participants must treat any additional bytes present as undefined values.)*

### 4.2.2 Order Rejected

`Order Rejected` messages are sent in response to a `New Order` which must be rejected. This message corresponds to a FIX Execution Report with *ExecType* (150) = 8 (Rejected). `Order Rejected` messages are unsequenced.

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

#### *4.2.2.1 OrderRejectedAUEquitiesV1*

| Field Name | Offset | Length | Data Type | Description |
|---|---|---|---|---|
| StartOfMessage | 0 | 2 | Binary | B0 E3 (58288) |
| MessageLength | 2 | 2 | Binary | 101 |
| MessageType | 4 | 2 | Binary | AE 0D (3502) |
| MatchingUnit | 6 | 1 | Binary | |
| Reserved | 7 | 1 | Binary | Value sent to participant not specified |
| SequenceNumber | 8 | 4 | Binary | *(Unsequenced: always 0)* |
| TransactionTime | 12 | 8 | DateTime | The time the event occurred in the Matching Engine (not the time the message was sent). |
| TradingParticipantID | 20 | 2 | Binary | The (4 digit) trading participant identifier. |
| ClOrdID | 22 | 20 | Text | Client order identifier of the rejected order. |
| OrderRejectReason | 42 | 1 | Text | See Section '4.4 - Order Reason Codes'. |
| Text | 43 | 60 | Text | Human readable text with more information. |

*(New fields may be introduced at the end of this message. Consequently, Participants must treat any additional bytes present as undefined values.)*

### 4.2.3 Order Modified

`Order Modified` messages are sent in response to a `Modify Order` to indicate that the order has been successfully modified.

In some cases, the last message to be received on an order's lifecycle will be an `Order Modified` message. The way to know the order is no longer live is to inspect *LeavesQty*. An example of this would be modification of an order while an execution is being generated, resulting in the order being reduced to zero outstanding quantity.

#### *4.2.3.1 OrderModifiedAUEquitiesV1*

| Field Name | Offset | Length | Data Type | Description |
|---|---|---|---|---|
| StartOfMessage | 0 | 2 | Binary | B0 E3 (58288) |
| MessageLength | 2 | 2 | Binary | 115 |
| MessageType | 4 | 2 | Binary | AF 0D (3503) |
| MatchingUnit | 6 | 1 | Binary | |
| Reserved | 7 | 1 | Binary | Value sent to participant not specified |
| SequenceNumber | 8 | 4 | Binary | *(Sequenced per MatchingUnit by Cboe Australia)* |
| TransactionTime | 12 | 8 | DateTime | The time the event occurred in the Matching Engine (not the time the message was sent). |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| TradingParticipantID | 20 | 2 | Binary | The (4 digit) trading participant identifier. |
| ClOrdID | 22 | 20 | Text | The *ClOrdID* of the modified order. |
| ExecInst | 42 | 1 | Text | The execution instruction of the modified order. |
| OrderQty | 43 | 4 | Binary | Order quantity. System limit is 999,999,999. |
| OrigClOrdID | 47 | 20 | Text | The *ClOrdID* of the original order. |
| Price | 67 | 8 | BinaryPrice | The limit or peg price of the order after modification. |
| MinQty | 75 | 4 | Binary | The minimum execution quantity after modification.<br>0 if no *MinQty* applies to the order. |
| MaxFloor | 79 | 4 | Binary | The maximum displayed value of the order after modification.<br>0 if no *MaxFloor* applies to the order. |
| LeavesQty | 83 | 4 | Binary | Quantity still open for further execution. If zero, the order is complete. |
| SecondaryOrderID | 87 | 8 | Binary | Cboe PITCH *OrderId* for the displayed portion of an iceberg order after a Modify Order increased the size of the iceberg order, otherwise 0. |
| ShortSellNakedQty | 95 | 4 | Binary | Copied from ModifyOrder |
| ShortSellCoveredQty | 99 | 4 | Binary | Copied from ModifyOrder |
| ShortSellLongQty | 103 | 4 | Binary | Copied from ModifyOrder |
| WorkingPrice | 107 | 8 | Binary | The Price at which an order is effectively working in the book. |
| BaseLiquidityIndicator | 115 | 1 | Alphanumeric | Indicates whether the trade added or removed liquidity.<br>A = Added Liquidity<br><br>R = Removed Liquidity - Indicates whether the trade added or removed liquidity.<br>A = Added Liquidity<br><br>R = Removed LiquidityIndicates whether the trade added or removed liquidity.<br><br>A = Added Liquidity<br>R = Removed LiquidityIndicates whether the trade added or removed liquidity.<br><br>A = Added Liquidity<br>R = Removed Liquidity |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| SubLiquidityIndicator | 116 | 1 | Text | Additional information about the liquidity of an order. Cboe Australia may add additional values without notice. Participants must gracefully ignore unknown values.<br>`NULL (0x00)` = No Additional Information<br>`I` = Added Hidden Liquidity that was price improved<br>`J` = Added Liquidity NBBO Joiner<br>`K` = Added Liquidity from the hidden (reserve) portion of an iceberg order<br>`S` = Added Liquidity NBBO Setter |

*(New fields may be introduced at the end of this message. Consequently, Participants must treat any additional bytes present as undefined values.)*

#### *4.2.3.2 OrderModifiedAUEquitiesV2*

| Field Name | Offset | Length | Data Type | Description |
|---|---|---|---|---|
| StartOfMessage | 0 | 2 | Binary | B0 E3 (58288) |
| MessageLength | 2 | 2 | Binary | 119 |
| MessageType | 4 | 2 | Binary | B9 0D (3513) |
| MatchingUnit | 6 | 1 | Binary | |
| Reserved | 7 | 1 | Binary | Value sent to participant not specified |
| SequenceNumber | 8 | 4 | Binary | *(Sequenced per MatchingUnit by Cboe Australia)* |
| TransactionTime | 12 | 8 | DateTime | The time the event occurred in the Matching Engine (not the time the message was sent). |
| TradingParticipantID | 20 | 2 | Binary | The (4 digit) trading participant identifier. |
| ClOrdID | 22 | 20 | Text | The *ClOrdID* of the modified order. |
| ExecInst | 42 | 1 | Text | The execution instruction of the modified order. |
| OrderQty | 43 | 4 | Binary | Order quantity. System limit is 999,999,999. |
| OrigClOrdID | 47 | 20 | Text | The *ClOrdID* of the original order. |
| Price | 67 | 8 | BinaryPrice | The limit or peg price of the order after modification. |
| MinQty | 75 | 4 | Binary | The minimum execution quantity after modification.<br>0 if no *MinQty* applies to the order. |
| MaxFloor | 79 | 4 | Binary | The maximum displayed value of the order after modification. |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| | | | | 0 if no *MaxFloor* applies to the order. |
| LeavesQty | 83 | 4 | Binary | Quantity still open for further execution. If zero, the order is complete. |
| SecondaryOrderID | 87 | 8 | Binary | Cboe PITCH *OrderId* for the displayed portion of an iceberg order after a Modify Order increased the size of the iceberg order, otherwise 0. |
| ShortSellNakedQty | 95 | 4 | Binary | Copied from ModifyOrder |
| ShortSellCoveredQty | 99 | 4 | Binary | Copied from ModifyOrder |
| ShortSellLongQty | 103 | 4 | Binary | Copied from ModifyOrder |
| WorkingPrice | 107 | 8 | Binary | The Price at which an order is effectively working in the book. |
| BaseLiquidityIndicator | 115 | 1 | Alphanumeric | `Indicates whether the trade added or removed liquidity.`<br>A = `Added Liquidity`<br><br>R = `Removed Liquidity - Indicates whether the trade added or removed liquidity.`<br>A = `Added Liquidity`<br><br>R = `Removed Liquidity` |
| SubLiquidityIndicator | 116 | 1 | Text | Additional information about the liquidity of an order. Cboe Australia may add additional values without notice. Participants must gracefully ignore unknown values.<br>`NULL (0x00)` = No Additional Information<br>`I` = Added Hidden Liquidity that was price improved<br>`J` = Added Liquidity NBBO Joiner<br>`K` = Added Liquidity from the hidden (reserve) portion of an iceberg order<br>`S` = Added Liquidity NBBO Setter |
| ExpireDate | 117 | 4 | Date | Date of order expiration (last day the order can trade) if *TimeInForce* is:<br>• 1 (GTC) or<br>• 6 (GTD) and *ExpireDate* specified. |

*(New fields may be introduced at the end of this message. Consequently, Participants must treat any additional bytes present as undefined values.)*

### 4.2.4 Modify Rejected

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

`Modify Rejected` messages are sent in response to a `Modify Order` for an order which cannot be modified. `Modify Rejected` messages are unsequenced.

This message corresponds to a FIX Execution Report with *MsgType* (35) = 9 (Order Cancel Reject) and *CxlRejResponseTo* (434) = 2 (Order Cancel/Replace Request).

#### *4.2.4.1 ModifyRejectedAUEquitiesV1*

| Field Name | Offset | Length | Data Type | Description |
|---|---|---|---|---|
| StartOfMessage | 0 | 2 | Binary | B0 E3 (58288) |
| MessageLength | 2 | 2 | Binary | 121 |
| MessageType | 4 | 2 | Binary | B0 0D (3504) |
| MatchingUnit | 6 | 1 | Binary | |
| Reserved | 7 | 1 | Binary | Value sent to participant not specified |
| SequenceNumber | 8 | 4 | Binary | *(Unsequenced: always 0)* |
| TransactionTime | 12 | 8 | DateTime | The time the event occurred in the Matching Engine (not the time the message was sent). |
| TradingParticipantID | 20 | 2 | Binary | The (4 digit) trading participant identifier. |
| ClOrdID | 22 | 20 | Text | ClOrdID of the `Modify Order`. |
| OrigClOrdID | 42 | 20 | Text | The *ClOrdID* of the original order. |
| ModifyRejectReason | 62 | 1 | Text | See Section '4.4 - Order Reason Codes'. |
| Text | 63 | 60 | Text | Human readable text with more information. |

*(New fields may be introduced at the end of this message. Consequently, Participants must treat any additional bytes present as undefined values.)*

### 4.2.5 Order Restated

When an Iceberg order reloads displayed quantity will receive an order restatement. If an order is cancelled or reduced due to Self-Trade Protection instructions (see NoTradeFeat), the order will be restated with reductions to *OrderQty* and/or *LeavesQty*. Note that *LeavesQty* may be reduced to 0, indicating that the order has effectively been cancelled for Self-Trade Protection.

#### *4.2.5.1 OrderRestatedAUEquitiesV1*

| Field Name | Offset | Length | Data Type | Description |
|---|---|---|---|---|
| StartOfMessage | 0 | 2 | Binary | B0 E3 (58288) |
| MessageLength | 2 | 2 | Binary | 86 |
| MessageType | 4 | 2 | Binary | B1 0D (3505) |
| MatchingUnit | 6 | 1 | Binary | |
| Reserved | 7 | 1 | Binary | Value sent to participant not specified |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| SequenceNumber | 8 | 4 | Binary | *(Sequenced per MatchingUnit by Cboe Australia)* |
|---|---|---|---|---|
| TransactionTime | 12 | 8 | DateTime | The time the event occurred in the Matching Engine (not the time the message was sent). |
| TradingParticipantID | 20 | 2 | Binary | The (4 digit) trading participant identifier. |
| ClOrdID | 22 | 20 | Text | The Client Order ID of the order being restated. |
| OrderID | 42 | 8 | Binary | Order identifier supplied by Cboe Australia. This identifier corresponds to the identifiers used in Cboe Australia market data products. |
| OrderQty | 50 | 4 | Binary | Order quantity. This may be less than the original OrderQty on the order. |
| Price | 54 | 8 | BinaryPrice | The Limit or Cap Price of the order.<br>0 for MOC. |
| LeavesQty | 62 | 4 | Binary | Quantity still open for further execution. If zero, the order is complete. |
| SecondaryOrderID | 66 | 8 | Binary | Cboe *OrderId* of contra side of prevented match, or Cboe PITCH *OrderId* for displayed portion of iceberg order after reloading. |
| LastPx | 74 | 8 | BinaryPrice | If *RestatementReason*=W, this is the price of the execution avoided by Self-Trade Protection.<br>Equivalent to *PreventedTradePx* (7903) in FIX.<br>Otherwise, zero. |
| LastShares | 82 | 4 | Binary | If *RestatementReason*=W, this is the size of the execution avoided by Self-Trade Protection.<br>Equivalent to *PreventedTradeLiquidityIndicator* (7905) in FIX.<br>Otherwise, zero. |
| BaseLiquidityIndicator | 86 | 1 | Text | For the trade avoided by self-trade, indicates whether the trade would have added or removed order liquidity.<br>Equivalent to *PreventedTradePx* (7903) in FIX.<br>`A = Added Liquidity`<br>`R = Removed Liquidity`<br>`NULL (0x00) = n/a` |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| RestatementReason | 87 | 1 | Text | The reason for this Order Restated message.<br>r = Iceberg order reload<br>W = Would have self-traded<br>R = Replaced by another Port (i.e. Purge-Amend Port) |

### 4.2.6 Order Execution

An `Order Execution` is sent for each fill on an order.

Rather than returning a monetary value indicating the rebate or charge for an execution, the *FeeCode* is an indication of a fee classification corresponding to an item on the venue's fee schedule.

#### *4.2.6.1 OrderExecutionAUEquitiesV1*

| Field Name | Offset | Length | Data Type | Description |
|---|---|---|---|---|
| StartOfMessage | 0 | 2 | Binary | B0 E3 (58288) |
| MessageLength | 2 | 2 | Binary | 114 |
| MessageType | 4 | 2 | Binary | B2 0D (3506) |
| MatchingUnit | 6 | 1 | Binary | |
| Reserved | 7 | 1 | Binary | Value sent to participant not specified |
| SequenceNumber | 8 | 4 | Binary | *(Sequenced per MatchingUnit by Cboe Australia)* |
| TransactionTime | 12 | 8 | DateTime | The time the event occurred in the Matching Engine (not the time the message was sent). |
| TradingParticipantID | 20 | 2 | Binary | The (4 digit) trading participant identifier. |
| ClOrdID | 22 | 20 | Text | The Client Order ID of the traded order. |
| ExecID | 42 | 8 | Binary | Execution Id. Unique across all matching units on a given day. Note: ExecIDs will be represented on ODROP and FIXDROP ports as base 36 ASCII. |
| LastCapacity | 50 | 1 | Text | Capacity of the order.<br>1 = Agent<br>2 = Cross as Agent<br>3 = Cross as Principal<br>4 = Principal<br>5 = Mixed<br>6 = Cross as Mixed<br>Values of 2, 3, or 6 indicate that the client has executed against themselves. Crossing is determined by reference to the first 3 digits of the trading participant ID. |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| LastMkt | 51 | 4 | Alphanumeric | Supported MIC code values are as follows:<br>CXAC = Limit<br>CXAP = Mid-Point<br>CXAN = Near Point<br>CXAF = Far Point<br>CXAM = MOC<br>**Note:** This value is determined by the venue of the resting or "Maker" order in a trade pair. Both sides of the trade contain the same value. |
| LastPx | 55 | 8 | BinaryPrice | Price of this fill.<br>For MOC Order executions, this price is the indicative price of the fill. |
| LastShares | 63 | 4 | Binary | Executed quantity. |
| Side | 67 | 1 | Text | 1 = Buy<br>2 = Sell<br>5 = Sell Short |
| Symbol | 68 | 6 | Alphanumeric | Instrument symbol (upper case alphanumeric) |
| LeavesQty | 74 | 4 | Binary | Quantity still open for further execution. If zero, the order is complete. |
| SecondaryOrderID | 78 | 8 | Binary | Denotes an alternative *OrderID* which is present on Cboe market data feeds (for example, to hide that a reserve (iceberg) order has reloaded). Otherwise, SecondaryOrderID will be 0. |
| ClearingParticipantID | 86 | 4 | Binary | The CHESS clearing participant identifier. Must be a valid 5-digit CHESS PID. |
| PriceImprovement | 90 | 8 | BinaryPrice | Amount of price improvement against the primary best bid/offer at the time of execution. Zero if no improvement was made or if there is no primary best bid or offer at the time of execution. |
| TransactID | 98 | 10 | Alphanumeric | Transaction Identifier as sent to CHESS for this order execution. |
| OnMarketCrossType | 108 | 1 | Text | Type of on-market crossing.<br>B = Trade was the result of a broker preferenced crossing<br>NULL (0x00) = other |
| TradeReportType | 109 | 1 | Text | X = Booking purpose trade.<br>NULL (0x00) = normal trade. |
| BaseLiquidityIndicator | 110 | 1 | Text | Indicates whether the trade added or removed liquidity.<br>A = Added Liquidity |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| | | | | R = Removed Liquidity<br>C = Auction / Uncrossing |
| SubLiquidityIndicator | 111 | 1 | Text | Additional information about the liquidity of an order. Cboe Australia may add additional values without notice. Participants must gracefully ignore unknown values.<br>NULL (0x00) = No Additional Information<br>I = Added Hidden Liquidity that was price improved<br>J = Added Liquidity NBBO Joiner<br>K = Added Liquidity from the hidden (reserve) portion of an iceberg order<br>S = Added Liquidity NBBO Setter<br>O = Opening Auction or intra-day auction trade<br>e = Closing Auction trade<br>h = Halt Auction (Re-Opening) trade |
| ContraParticipantID | 112 | 2 | Binary | Trading Participant Identifier (PID) of the counter party on a trade. "0" If unattributed. |
| FeeCode | 114 | 2 | Alphanumeric | Indicates fee associated with an execution. Fee codes are published in the pricing schedule. New fee codes may be sent with little notice. Participants are encouraged to code their systems to accept unknown fee codes. |

*(New fields may be introduced at the end of this message. Consequently, Participants must treat any additional bytes present as undefined values.)*

### 4.2.7 Order Cancelled

#### *4.2.7.1 OrderCancelledAUEquitiesV1*

| Field Name | Offset | Length | Data Type | Description |
|---|---|---|---|---|
| StartOfMessage | 0 | 2 | Binary | B0 E3 (58288) |
| MessageLength | 2 | 2 | Binary | 41 |
| MessageType | 4 | 2 | Binary | B3 0D (3507) |
| MatchingUnit | 6 | 1 | Binary | |
| Reserved | 7 | 1 | Binary | Value sent to participant not specified |
| SequenceNumber | 8 | 4 | Binary | *(Sequenced per MatchingUnit by Cboe Australia)* |
| TransactionTime | 12 | 8 | DateTime | The time the event occurred in the Matching Engine (not the time the message was sent). |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| TradingParticipantID | 20 | 2 | Binary | The (4 digit) trading participant identifier. |
| ClOrdID | 22 | 20 | Text | Client Order ID of the cancelled order. |
| CancelReason | 42 | 1 | Text | See Section '4.4 - Order Reason Codes'. |

*(New fields may be introduced at the end of this message. Consequently, Participants must treat any additional bytes present as undefined values.)*

### 4.2.8 Cancel Rejected

A `Cancel Rejected` message is sent in response to a `Cancel Order` message to indicate that the cancellation cannot occur. `Cancel Rejected` messages are unsequenced.

#### *4.2.8.1 CancelRejectedAUEquitiesV1*

| Field Name | Offset | Length | Data Type | Description |
|---|---|---|---|---|
| StartOfMessage | 0 | 2 | Binary | B0 E3 (58288) |
| MessageLength | 2 | 2 | Binary | 101 |
| MessageType | 4 | 2 | Binary | B4 0D (3508) |
| MatchingUnit | 6 | 1 | Binary | |
| Reserved | 7 | 1 | Binary | Value sent to participant not specified |
| SequenceNumber | 8 | 4 | Binary | *(Unsequenced: always 0)* |
| TransactionTime | 12 | 8 | DateTime | The time the event occurred in the Matching Engine (not the time the message was sent). |
| TradingParticipantID | 20 | 2 | Binary | The (4 digit) trading participant identifier. |
| ClOrdID | 22 | 20 | Text | The Client Order ID of the `Cancel Order` message. |
| CancelRejectReason | 42 | 1 | Text | See Section '4.4 - Order Reason Codes'. |
| Text | 43 | 60 | Text | Human readable text with more information. |

*(New fields may be introduced at the end of this message. Consequently, Participants must treat any additional bytes present as undefined values.)*

### 4.2.9 Order Execution Restated

Order Execution Restated messages will be issued for all (unbroken) fills of MOC orders after the closing price has been determined. The *LastPx* value will inform the Participant of the final trade price for the MOC transaction.

#### *4.2.9.1 ExecutionRestatementAUEquitiesV1*

| Field Name | Offset | Length | Data Type | Description |
|---|---|---|---|---|
| StartOfMessage | 0 | 2 | Binary | B0 E3 (58288) |
| MessageLength | 2 | 2 | Binary | 83 |
| MessageType | 4 | 2 | Binary | B5 0D (3509) |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| MatchingUnit | 6 | 1 | Binary | |
| Reserved | 7 | 1 | Binary | Value sent to participant not specified |
| SequenceNumber | 8 | 4 | Binary | *(Sequenced per MatchingUnit by Cboe Australia)* |
| TransactionTime | 12 | 8 | DateTime | The time the event occurred in the Matching Engine (not the time the message was sent). |
| TradingParticipantID | 20 | 2 | Binary | The (4 digit) trading participant identifier. |
| ClOrdID | 22 | 20 | Text | The Client Order ID of the traded order. |
| ExecID | 42 | 8 | Binary | Execution Id. Unique across all matching units on a given day. Note: ExecIDs will be represented on ODROP and FIXDROP ports as base 36 ASCII. |
| LastPx | 50 | 8 | BinaryPrice | Price of this fill. |
| LastShares | 58 | 4 | Binary | Executed quantity. |
| Side | 62 | 1 | Text | 1 = Buy<br>2 = Sell<br>5 = Sell Short |
| Symbol | 63 | 6 | Alphanumeric | Instrument symbol (upper case alphanumeric) |
| TransactID | 69 | 10 | Alphanumeric | Transaction Identifier as sent to CHESS for this order execution. |
| BaseLiquidityIndicator | 79 | 1 | Text | Indicates whether the trade added or removed liquidity.<br>A = Added Liquidity<br>R = Removed Liquidity |
| ContraParticipantID | 80 | 2 | Binary | Trading Participant Identifier (PID) of the counter party on a trade. "0" If unattributed. |
| FeeCode | 82 | 2 | Alphanumeric | Indicates fee associated with an execution. Fee codes are published in the pricing schedule. New fee codes may be sent with little notice. Participants are encouraged to code their systems to accept unknown fee codes. |
| ExecRestatementReason | 84 | 1 | Text | The reason for this Order Execution Restated message.<br>R = Market on close re-pricing. |

*(New fields may be introduced at the end of this message. Consequently, Participants must treat any additional bytes present as undefined values.)*

### 4.2.10 Purge Acknowledgement

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

A `Purge Acknowledgment` is an unsequenced message sent when a `Purge Orders` message requesting an order purge has completed cancelling all individual orders.

#### *4.2.10.1 PurgeAcknowledgementAUEquitiesV1*

| Field Name | Offset | Length | Data Type | Description |
|---|---|---|---|---|
| StartOfMessage | 0 | 2 | Binary | B0 E3 (58288) |
| MessageLength | 2 | 2 | Binary | 42 |
| MessageType | 4 | 2 | Binary | B6 0D (3510) |
| MatchingUnit | 6 | 1 | Binary | |
| Reserved | 7 | 1 | Binary | Value sent to participant not specified |
| SequenceNumber | 8 | 4 | Binary | *(Unsequenced: always 0)* |
| TransactionTime | 12 | 8 | DateTime | The time the event occurred in the Matching Engine (not the time the message was sent). |
| MassCancelID | 20 | 20 | Text | User-defined identifier of the mass cancel or purge request. |
| CancelledOrderCount | 40 | 4 | Binary | Number of orders cancelled. |

*(New fields may be introduced at the end of this message. Consequently, Participants must treat any additional bytes present as undefined values.)*

### 4.2.11 Purge Rejected

A `Purge Rejected` message is sent in response to a `Purge Orders` message to indicate that the mass cancellation cannot occur. `Purge Rejected` messages are unsequenced.

#### *4.2.11.1 PurgeRejectedAUEquitiesV1*

| Field Name | Offset | Length | Data Type | Description |
|---|---|---|---|---|
| StartOfMessage | 0 | 2 | Binary | B0 E3 (58288) |
| MessageLength | 2 | 2 | Binary | 99 |
| MessageType | 4 | 2 | Binary | B7 0D (3511) |
| MatchingUnit | 6 | 1 | Binary | |
| Reserved | 7 | 1 | Binary | Value sent to participant not specified |
| SequenceNumber | 8 | 4 | Binary | *(Unsequenced: always 0)* |
| TransactionTime | 12 | 8 | DateTime | The time the event occurred in the Matching Engine (not the time the message was sent). |
| MassCancelID | 20 | 20 | Text | User-defined identifier of the mass cancel or purge request. |
| PurgeRejectReason | 40 | 1 | Text | Reason for the purge rejection. See Section '4.4 - Order Reason Codes'. |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| Text | 41 | 60 | Text | Human readable text with more information. |

*(New fields may be introduced at the end of this message. Consequently, Participants must treat any additional bytes present as undefined values.)*

### 4.2.12 Trade Cancel or Correct

Used to provide notification that a trade has been cancelled (busted) or corrected (price change only). The *CorrectedPrice* field will be set to 0 for cancelled trades and to the new trade price for corrected trades (note that currently Cboe Australia only supports trade cancellations, therefore *CorrectedPrice* will always be 0).

#### *4.2.12.1 TradeCancelCorrectAUEquitiesV1*

| Field Name | Offset | Length | Data Type | Description |
|---|---|---|---|---|
| StartOfMessage | 0 | 2 | Binary | B0 E3 (58288) |
| MessageLength | 2 | 2 | Binary | 98 |
| MessageType | 4 | 2 | Binary | B8 0D (3512) |
| MatchingUnit | 6 | 1 | Binary | |
| Reserved | 7 | 1 | Binary | Value sent to participant not specified |
| SequenceNumber | 8 | 4 | Binary | *(Sequenced per MatchingUnit by Cboe Australia)* |
| TransactionTime | 12 | 8 | DateTime | The time the event occurred in the Matching Engine (not the time the message was sent). |
| TradingParticipantID | 20 | 2 | Binary | The (4 digit) trading participant identifier. |
| ClOrdID | 22 | 20 | Text | Client Order ID of the traded order |
| ExecRefID | 42 | 8 | Binary | Refers to the *ExecId* of the fill being cancelled or corrected. |
| LastPx | 50 | 8 | BinaryPrice | Price of this fill. |
| LastShares | 58 | 4 | Binary | Executed quantity. |
| OrigTime | 62 | 8 | DateTime | *TransactionTime* of the original trade. |
| Side | 70 | 1 | Text | 1 = Buy<br>2 = Sell<br>5 = Sell Short |
| Symbol | 71 | 6 | Alphanumeric | Instrument symbol (upper case alphanumeric) |
| ClearingParticipantID | 77 | 4 | Binary | The CHESS clearing participant identifier. Must be a valid 5-digit CHESS PID. |
| TransactID | 81 | 10 | Alphanumeric | Transaction Identifier as sent to CHESS for this order execution. |
| CorrectedPrice | 91 | 8 | BinaryPrice | The corrected price of the trade, or 0 for a trade cancellation. |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| BaseLiquidityIndicator | 99 | 1 | Text | Indicates whether the trade added or removed liquidity.<br>A = Added Liquidity<br>R = Removed Liquidity<br>C = Auction / Uncrossing. |

*(New fields may be introduced at the end of this message. Consequently, Participants must treat any additional bytes present as undefined values.)*

### 4.2.13 Carried Restatement

The Carried Restatement message is used to indicate the restoration of a multi-day GTC or GTD orders at the start of a new business day.

#### *4.2.13.1 CarriedRestatementAUEquitiesV1*

| Field Name | Offset | Length | Data Type | Description |
|---|---|---|---|---|
| StartOfMessage | 0 | 2 | Binary | B0 E3 (58288) |
| MessageLength | 2 | 2 | Binary | 192 |
| MessageType | 4 | 2 | Binary | BA 0D (3514) |
| MatchingUnit | 6 | 1 | Binary | |
| Reserved | 7 | 1 | Binary | Value sent to participant not specified |
| SequenceNumber | 8 | 4 | Binary | *(Sequenced per MatchingUnit by Cboe Australia)* |
| TransactionTime | 12 | 8 | DateTime | The time the event occurred in the Matching Engine (not the time the message was sent). |
| Account | 20 | 10 | Text | (Copied from order) |
| TradingParticipantID | 30 | 2 | Binary | (Copied from order / port default) |
| ClOrdID | 32 | 20 | Text | (Copied from order) |
| ExecInst | 52 | 1 | Text | (Copied from order) |
| OrderID | 53 | 8 | Binary | Order identifier supplied by Cboe Australia. This identifier corresponds to the identifiers used in Cboe Australia market data products. |
| OrderQty | 61 | 4 | Binary | (Copied from order) |
| OrdType | 65 | 1 | Text | (Copied from order) |
| Price | 66 | 8 | BinaryPrice | (Copied from order) |
| Capacity | 74 | 1 | Text | (Copied from order) |
| Side | 75 | 1 | Text | (Copied from order) |
| Symbol | 76 | 6 | Alphanumeric | (Copied from order) |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| TimeInForce | 82 | 1 | Text | (Copied from order) |
| MinQty | 83 | 4 | Binary | (Copied from order) |
| MaxFloor | 87 | 4 | Binary | (Copied from order) |
| ExpireTime | 91 | 8 | DateTime | (Copied from order) |
| ClearingParticipantID | 99 | 4 | Binary | (Copied from order) |
| ClearingAccount | 103 | 10 | Alphanumeric | (Copied from order) |
| Undisclosed | 113 | 1 | Boolean | (Copied from order) |
| ClientCrossRef | 114 | 15 | Text | (Copied from order) |
| ShortSellNakedQty | 129 | 4 | Binary | (Copied from order) |
| ShortSellCoveredQty | 133 | 4 | Binary | (Copied from order) |
| ShortSellLongQty | 137 | 4 | Binary | (Copied from order) |
| RiskGroupID | 141 | 2 | Binary | (Copied from order) |
| NoTradeFeat | 143 | 1 | Text | (Copied from order / port default) |
| NoTradeKey | 144 | 10 | Text | (Copied from order) |
| MEQSE | 154 | 1 | Text | (Copied from order) |
| OriginOfTrans | 155 | 20 | Text | (Copied from order) |
| IntermediaryID | 175 | 10 | Text | (Copied from order) |
| DirectedWholesaleIndicator | 185 | 1 | Boolean | (Copied from order) |
| RemoveCrossingFromClearing | 186 | 1 | Boolean | (Copied from order) |
| BrokerPreferencing | 187 | 1 | Boolean | (Copied from order) |
| RoutingInst | 188 | 1 | Text | (Copied from order) |
| DisplayIndicator | 189 | 1 | Text | (Copied from order) |
| ExpireDate | 190 | 4 | Date | Date of order expiration (last day the order can trade) if *TimeInForce* is:<br>• 1 (GTC) or<br>• 6 (GTD) and *ExpireDate* specified. |

*(New fields may be introduced at the end of this message. Consequently, Participants must treat any additional bytes present as undefined values.)*

### 4.2.14 Done For Day

Done For Day messages in BOE represent orders persisted during the current day session to carry over to the next session and will be sent after the end of trading for the associated product and before system recycle.

#### *4.2.14.1 DoneForDayAUEquitiesV1*

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| Field Name | Offset | Length | Data Type | Description |
|---|---|---|---|---|
| StartOfMessage | 0 | 2 | Binary | B0 E3 (58288) |
| MessageLength | 2 | 2 | Binary | 44 |
| MessageType | 4 | 2 | Binary | BB 0D (3515) |
| MatchingUnit | 6 | 1 | Binary | |
| Reserved | 7 | 1 | Binary | Value sent to participant not specified |
| SequenceNumber | 8 | 4 | Binary | *(Sequenced per MatchingUnit by Cboe Australia)* |
| TransactionTime | 12 | 8 | DateTime | The time the event occurred in the Matching Engine (not the time the message was sent). |
| ClOrdID | 20 | 20 | Text | Client Order ID of the traded order |
| Symbol | 40 | 6 | Alphanumeric | Instrument symbol (upper case alphanumeric) |

## 4.3 Application Message Fields

| Field Name | FIX Field Name | Length | Data Type | Description |
|---|---|---|---|---|
| **Account** | 1 | 10 | Text | Used by a trading participant to provide a broker cross reference.<br>Valid characters are ASCII 32-126. |
| **BaseLiquidityIndicator** | 9730 | 1 | Alphanumeric | Indicates whether the trade added or removed liquidity.<br>A = Added Liquidity<br>R = Removed Liquidity<br>C = Auction / Uncrossing |
| **BrokerPreferencing** | 8182 | 1 | Text | Y = match with other orders using broker preferencing when possible.<br>N = do not apply broker preferencing<br>NULL (0x00) = apply port default |
| **CancelledOrderCount** | 7696 | 4 | Binary | Number of orders cancelled. |
| **CancelOrigOnReject** | 9619 | 1 | Alpha | Y = Cancel original order if modification fails. |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| | | | | N = Leave original order alone.<br>NULL (0x00) = Apply port default |
| **CancelReason** | 58** | 1 | Text | See Section '4.4 - Order Reason Codes'. |
| **CancelRejectReason** | 58** | 1 | Text | See Section '4.4 - Order Reason Codes'. |
| **Capacity** | 47 | 1 | Alphanumeric | The capacity for the order. Required on all orders.<br>A = Agency<br>P = Principal<br>X = Mixed |
| **ClearingAccount** | 440 | 10 | Alphanumeric | Supplemental identifier.<br>Recorded and made available on order acknowledgements. Available via Drop feeds.<br>This field may be blank.<br>10 character alphanumeric. Spaces, symbols, or special characters are not allowed.<br>This field is not sent to CHESS. |
| **ClearingParticipantID** | 439 | 4 | Binary | The CHESS clearing participant identifier. Must be a valid 5-digit CHESS PID. |
| **ClientCrossRef** | 6802 | 15 | Text | Can be used to provide additional participant-side cross reference information. |
| **ClOrdID** | 11 | 20 | Text | Unique Id chosen by the participant. Characters in the ASCII range 33-126 are allowed, except for **comma, semicolon, pipe, 'at' (@)**, and **double quotes (").**<br>If the ClOrdID matches a live order, the order will be rejected as duplicate. A leading **tilde** (~) cannot be sent on any *ClOrdID* and will result in a reject. These are reserved for internal use by Cboe Australia and could be received because of a Cboe-generated *ClOrdID*.<br>**Note:** Although Cboe Australia only enforces uniqueness of *ClOrdID* values among currently live orders, |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| | | | | using unique *ClOrdID* values is strongly recommended. |
| **ContraParticipantID** | 9862 | 2 | Binary | Trading Participant Identifier (PID) of the counter party on a trade. "0" If unattributed. |
| **CorrectedPrice** | 9620 | 8 | BinaryPrice | The corrected price of the trade, or 0 for a trade cancellation. |
| **DirectedWholesaleIndicator** | 8140 | 1 | Text | Market Integrity Rule defined field to identify wholesale participant.<br>Y = True<br>N = False |
| **DisplayIndicator** | 9479 | 1 | Text | Treatment of Post Only orders on entry if creating a locked NBBO.<br>V = Default. Post Only orders which would lock the NBBO on entry are allowed to do so.<br>R = Cancel back a Post Only order on entry if it were to create a locked NBBO.<br>All Post Only orders on entry which would create a crossed NBBO are cancelled back. |
| **ExecId** | 17 | 8 | Binary | Execution Id. Unique across all matching units on a given day. Note: ExecIDs will be represented on ODROP and FIXDROP ports as base 36 ASCII. * |
| **ExecInst** | 18 | 1 | Text | All values not defined below are rejected. Ignored if blank (0x00). |

* Example conversion:

| Decimal | Base 36 |
|---|---|
| 28294005440239 | A1234B567 |
| 76335905726621 | R248BC23H |
| 728557228187 | 09AP05V2Z |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| | | | | P = Market Peg (peg Buy [Sell] to NBBO Offer [Bid], one tick within NBBO)<br>R = Primary Peg (peg Buy [Sell] to NBBO Bid [Offer], one tick within NBBO)<br>M = Midpoint (peg to NBBO Midpoint)<br>j = BestX Order (insert the limit order at the best visible price on same side). Not applicable to IOC or FOK.<br>k = MarketX Order (insert the limit order against the best visible price on opposite side). |
| **ExecRefId** | 19 | 8 | Binary | Refers to the *ExecId* of the fill being cancelled or corrected. |
| **ExecRestatementReason** | n/a | 1 | Text | The reason for this Order Execution Restated message.<br>R = Market on close re-pricing. |
| **ExpireTime** | 424 | 8 | DateTime | Required for GTD order (*TimeInForce* = 6), *ExpireTime* specifies the date-time (in UTC) when the order expires. GTD orders will expire at the earlier of the end of the current trading day and the *ExpireTime*. |
| **ExpireDate** | 432 | 4 | Date | Only applicable for *TimeInForce* (59) = '6' (GTD) orders.<br>Specifies the date on which a GTD order will expire using format "YYYYMMDD". Orders will expire at the end of the *ExpireDate*.<br>Ignored if the date is on or before the current date.<br>Required if *TimeInForce* is 6 (GTD) and *ExpireTime* is not specified.<br>Ignored if *TimeInForce* is not 6 (GTD) or if *ExpireTime* is present. |
| **FeeCode** | 9882 | 2 | Alphanumeric | Indicates fee associated with an execution. Fee codes are published in the pricing schedule. New fee codes may be sent with little notice. Participants are encouraged to code |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| | | | | their systems to accept unknown fee codes. |
| **IntermediaryID** | 8139 | 10 | Text | Market Integrity Rule defined field for Intermediary Identifier–may be populated with AFS license number. Characters in ASCII range 32-126 are allowed. |
| **LastCapacity** | 29 | 1 | Text | Capacity of the order.<br>1 = Agent<br>2 = Cross as Agent<br>3 = Cross as Principal<br>4 = Principal<br>5 = Mixed<br>6 = Cross as Mixed<br>Values of 2, 3, or 6 indicate that the client has executed against themselves. Crossing is determined by reference to the first 3 digits of the trading participant ID. |
| **LastMkt** | 30 | 4 | Alphanumeric | Supported MIC code values are as follows:<br>CXAC = Limit<br>CXAP = Mid-Point<br>CXAN = Near Point<br>CXAF = Far Point<br>CXAM = MOC<br>**Note:** This value is determined by the venue of the resting or "Maker" order in a trade pair. Both sides of the trade contain the same value. |
| **LastPx** | 31 | 8 | BinaryPrice | Price of this fill. |
| **LastShares** | 32 | 4 | Binary | Executed quantity. |
| **LeavesQty** | 151 | 4 | Binary | Quantity still open for further execution. If zero, the order is complete. |
| **MassCancelID** | 7695 | 20 | Text | User-defined identifier of the mass cancel or purge request. |
| **MassCancelInst** | 7700 | 16 | Text | At least one character must be provided (PID Filter). Contiguous characters must be specified up to total length. Truncated (unspecified) |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| | | | | characters will default to values indicated below.<br>***1st Character : PID Filter***<br>A = No filtering by PID is performed.<br>F = All orders that were sent under the PID specified in *TradingParticipantID* will be cancelled.<br>***2nd Character : Acknowledgement Style***<br>M = (default) Individual Execution Reports are sent for each cancelled order.<br>S = Single `Purge Acknowledgement` sent once all cancels have been processed. Single `Purge Acknowledgement` will contain *MassCancelId* and *CancelledOrderCount*. *MassCancelId* must be specified, or the `Purge Request` will be rejected.<br>B = Both individual `Execution Reports` and single `Purge Acknowledgement` will be sent. Also requires *MassCancelId* to be specified or the `Purge Request` will be rejected.<br>***3rd Character : Order Scope***<br>D = (default) Purge Day orders<br>M = Purge GTC and GTD orders |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| **MaxFloor** | 111 | 4 | Binary | Portion of *OrderQty* to display. Used for iceberg orders and hidden orders.<br><br>Limit orders (*OrdType* = 2) having *MaxFloor* > 0 are iceberg orders. For iceberg orders, *MaxFloor* is the displayed quantity. The balance is reserve. The displayed quantity of each order at a price level is decremented first.<br><br>Limit orders (*OrdType*=2) having *MaxFloor* = 0 will display their full size.<br><br>Pegged orders (*OrdType* = P) and MOC orders (*OrdType* = 5) must use *MaxFloor* =0, which hides the entire quantity.<br><br>**Ignored** if order is IOC or FOK. |
| **MEQSE** | 8081 | 1 | Text | If true, then any single fill will be for at least *MinQty* shares.<br><br>Y = True<br>N = False<br><br>If True, then the order will be rejected unless *MinQty* is also specified or the *OrdType* is not 5 or P (MOC or Pegged). |
| **MinQty** | 110 | 4 | Binary | Minimum fill quantity for orders. Required if *MEQSE* is true. Accepted only on IOC, MOC, and Pegged orders.<br><br>If MEQSE is false, multiple consecutive fills may be created such that their combined size satisfies the *MinQty* requirement.<br><br>When the remaining size on an order is less than the defined *MinQty*, *LeavesQty* is treated as the *MinQty*. |
| **ModifyRejectReason** | 103 | 1 | Text | See Section '4.4 - Order Reason Codes'. |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| **NoTradeFeat** | 7713 | 1 | Alpha | Ignored if *NoTradeKey* is not set. Declares the handling of self-trading.<br>U = No self trade protection<br>N = Cancel Newest<br>O = Cancel Oldest<br>B = Cancel Both<br>S = Cancel Smallest<br>D = Decrement larger/Cancel Smaller<br>d = Same as D above, but only decrement *LeavesQty*. Do not restate *OrderQty*.<br>X = Booking Purpose Trade; the resultant trade is reported to CHESS as a booking purpose trade and as a cancel on public market data feeds.<br>NULL (0x00) = Apply the port default.<br>If the older and newer orders in a self-trade have conflicting instructions, the instruction of the newer order will be honoured, except that if the inbound order specifies Decrement and the resting order does not, and the resting order is larger, then both orders will be cancelled. This exception is to protect the order entry software for the resting order from receiving an unexpected restatement message. |
| **NoTradeKey** | 7714 | 10 | Text | Interactions of orders from the same participant (defined by reference to the first 3 digits of the trading PID) with the same *NoTradeKey* value will be managed by the newer order's *NoTradeFeat* instruction. |
| **OnMarketCrossType** | 8183 | 1 | Text | Type of on-market crossing.<br>B = Trade was the result of a broker preferenced crossing<br>NULL (0x00) = other |
| **OrderID** | 37 | 8 | Binary | Order identifier supplied by Cboe Australia. This identifier corresponds to the identifiers used in Cboe Australia market data products. |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| **OrderQty** | 31 | | Binary | Order quantity. System limit is 999,999,999. |
| **OrderRejectReason** | 103 | 1 | Text | See Section '4.4 - Order Reason Codes'. |
| **OrdType** | 40 | 1 | Alphanumeric | 2 = Limit<br>5 = Market on Close (MOC)<br>P = Pegged<br>MOC requires *MaxFloor* to be 0 if *TimeInForce* is 0 or 6.<br>Pegged requires *ExecInst* to be P, R, or M, and requires *MaxFloor* to be 0. |
| **OrigClOrdID** | 41 | 20 | Text | The *ClOrdID* of the original order. |
| **OriginOfTrans** | 8138 | 20 | Text | Market Integrity Rule defined field for Origin of Order (ex. ACN, ABN, CHESS HIN). |
| **OrigTime** | 42 | 8 | DateTime | *TransactionTime* of the original trade. |
| **Price** | 44 | 8 | BinaryPrice | If *OrdType* is 2 (limit), *Price* must be an applicable minimum trading increment.<br>If *OrdType* is P (peg), *Price* must specify a cap price as an applicable minimum trading increment, or a ½ increment if *ExecInst*=M (mid).<br>If *OrdType* is 5 (MOC), *Price* must be 0. |
| **PriceImprovement** | 639 | 8 | BinaryPrice | Amount of price improvement against the primary best bid/offer at the time of execution. Zero if no improvement was made or if there is no primary best bid or offer at the time of execution. |
| **PurgeRejectReason** | 58** | 1 | Text | Reason for the purge rejection. See Section '4.4 - Order Reason Codes'. |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| **RemoveCrossingFromClearing** | 8177 | 1 | Text | If "Y", instructs Cboe Australia on how to process crossed trades with CHESS. Please contact Cboe Australia to confirm settings.<br>`Y` = True<br>`N` = False |
| **RestatementReason** | n/a | 1 | Text | The reason for this Order Restated message.<br>`r` = Iceberg order reload<br>`W` = Would have self-traded |
| **RiskGroupID** | 7699 | 2 | Binary | Used to group orders for use in order purges where multiple orders can be cancelled by specifying a list of *RiskGroupID*s. |
| **RiskGroupIdCnt** | 7698 | 1 | Binary | Number of repeating *RiskGroupID* included in this message. Value 0 to 10. |
| **RoutingInst** | 9303 | 1 | Text | Indicates the order is a Post Only order.<br>`P` = Post Only (never take liquidity) |
| **SecondaryOrderID** | 198 | 8 | Binary | Present on a Self-Trade Protection triggered restatement, on an iceberg reload restatement, and on an Order Modified acknowledgement that increases the size of an iceberg order.<br>Cboe *OrderId* of contra side of prevented match, or Cboe PITCH *OrderId* for displayed portion of an iceberg order after reload, or Cboe PITCH *OrderId* for the displayed portion of an iceberg order after a `Modify Order` increased the size of the iceberg order. |
| **ShortSellNakedQty** | 6803 | 4 | Binary | Defines the portion of O*rderQty* (38) that is naked. |
| **ShortSellCoveredQty** | 6804 | 4 | Binary | Defines the portion of *OrderQty* (38) that is covered. |
| **ShortSellLongQty** | 6805 | 4 | Binary | Defines the portion of *OrderQty* (38) that is owned by the participant who placed the order.<br>Orders having *Side* = 5 (sell short) will be rejected unless |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| | | | | *ShortSellNakedQty* + *ShortSellCoveredQty* + *ShortSellLongQty* is equal to the *OrderQty* and *ShortSellLongQty* < *OrderQty*. |
| **Side** | 54 | 1 | Text | 1 = Buy<br>2 = Sell<br>5 = Sell Short |
| **SourceMatchingUnit** | - | 1 | Binary | The matching unit reporting the purge. |
| **SubLiquidityIndicator** | 9730* | 1 | Text | Additional information about the liquidity of an order. Cboe Australia may add additional values without notice. Participants must gracefully ignore unknown values.<br>NULL (0x00) = No Additional Information<br>I = Added Hidden Liquidity that was price improved<br>J = Added Liquidity NBBO Joiner<br>K = Added Liquidity from the hidden (reserve) portion of an iceberg order<br>S = Added Liquidity NBBO Setter<br>O = Opening Auction or intra-day auction trade.<br>e = Closing Auction trade.<br>h = Halt Auction (Re-Opening) |
| **Symbol** | 55 | 6 | Alphanumeric | Instrument symbol (upper case alphanumeric) |
| **TargetMatchingUnit** | - | 1 | Binary | Matching unit to which the purge should be restricted. Zero is considered "all matching engines". |
| **Text** | 58 | 60 | Text | Human readable text with more information. |
| **TimeInForce** | 59 | 1 | Text | 0 = DAY (default)<br>1 = GTC (Good Till Cancel).<br>3 = IOC (portion not filled immediately is cancelled).<br>4 = FOK (an IOC where the entire size must be filled immediately, otherwise the order will be cancelled back).<br>6 = GTD. Expires at the specified: |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | | | |
|---|---|---|---|---|
| | | | | • *ExpireTime* or end of day, or<br>• *ExpireDate* for multi-day orders. |
| **TradeReportType** | | | | X = Booking purpose trade.<br>NULL (0x00) = normal trade. |
| **TradingParticipantID** | 115 | 2 | Binary | The (4 digit) trading participant identifier. |
| **TransactID** | 6807 | 10 | Alphanumeric | Transaction Identifier as sent to CHESS for this order execution. |
| **TransactionTime** | 60 | 8 | DateTime | The time the event occurred in the Matching Engine (not the time the message was sent). |
| **Undisclosed** | 6801 | 1 | Text | If true, indicates an undisclosed order.<br>Y = True<br>N = False |
| **WorkingPrice** | 9690 | 8 | BinaryPrice | The Price at which an order is effectively working in the book. |

** In FIX, the first character of Tag 58 (Text) provides the reason code related to the subsequent human-readable text.

## 4.4 Order Reason Codes

The following is a list of all reason codes used by Cboe Australia. These reason codes are used in a variety of contexts (order cancellations and order rejections). All reasons are not valid in all contexts. The reason code will be followed by free-form text. The specific text the system delivers may vary from the text listed below to provide clarification of the reject reason. Cboe Australia may add additional reason codes without notice. Participants must gracefully ignore unknown values.

A = Admin
- Most rejects due to field validation errors will be reported with a reason of Admin.

D = Duplicate identifier (e.g., ClOrdID)
- Duplicate *ClOrdID* detected. Note that Cboe Australia will not detect every possible occurrence of a duplicate by a participant.

H = Halted

I = Incorrect data centre
- Message directed to the DR data centre

K = Order rate threshold exceeded

L = Order would lock or cross NBBO

N = Ran out of liquidity to execute against
- Cancel reason for the unexecuted size of an IOC/FOK order.

O = *ClOrdID* does not match a known order

U = User requested

W = Add liquidity only order would remove

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

X = Order expired
- Cancellation of an expired GTD order

Y = Symbol not supported
Z = Unforeseen reason
o = Max open orders count exceeded
y = Order received by Cboe Australia during replay
z = Session end

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

# 5. Port Attributes

| Attribute | Default | Description |
|---|---|---|
| Allowed Trading PID(s) | Port Owner's PID | Trading PID(s) allowed for the port. |
| Enable MOCs | Yes | Allow or disallow Market-On-Close (MOC) order type. |
| Enable Pegs | Yes | Allow or disallow all pegged orders. |
| Enable Primary Pegs | Yes | Allow or disallow primary peg orders. |
| Enable Market Pegs | Yes | Allow or disallow market peg orders. |
| Enable Midpoint Pegs | Yes | Allow or disallow midpoint peg orders. |
| Enable Broker Preference | No | Allow or disallow broker preferencing. |
| Allow Test Symbols Only | No | Allow or disallow orders in non-test symbols. |
| Maximum Notional Value Per Order | Unlimited | Configurable maximum notional dollar value allowed on any individual order. |
| Maximum Order Size | 999,999,999 | Maximum number of shares allowed per order. |
| Cancel on ME Disconnect | No | When set to "No", this setting allows orders to remain open on a Matching Unit failover. When set to "Yes", all open orders associated with a session are immediately cancelled in the event of loss of connectivity to a Matching Unit. If a failover takes longer than five minutes, then all orders are cancelled unconditionally.<br>When set to "Day", Cancel Day orders only and do not cancel GTC or GTD orders, in the event of a Matching Unit failover. |
| Cancel on Disconnect | *Option #1* | Cboe Australia will offer Participants two options for cancelling orders because of a session disconnect:<br>1. Cancel All Open Orders (includes Market-On-Close orders) (default)<br>2. Do Not Cancel Any Open Orders<br>3. Cancel Day orders only and do not cancel GTC or GTD orders. |
| Default *NoTradeFeat* | N (cancel newest) | Specifies default value for *NoTradeFeat* (7713). |
| Default Post Only | Blank | If set to "P", all eligible Lit orders to be Post Only.<br>Eligible includes:<br>• Limit (*OrdType* = 2), and<br>• *TimeInForce* = 0 (Day) or 1 (GTC) or 6 (GTD). |
| Default Post Only Lock Instruction | V | When a different default Post Only locked NBBO behaviour is desired for Post Only orders, this port attribute may be used. See *DisplayIndicator*. |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | |
|---|---|---|
| Cancel on Reject | Yes | Cancels an order upon a modify reject for that order. |
| Cancel on Halt | No | Cancel open orders for a symbol upon a halt. When set to “Day”, Cancel Day orders only and do not cancel GTC or GTD orders, in the event of a halt. |
| Reject Orders on Drop Port Disconnect | No | Allows a participant to associate one or more drop port(s) to an order entry port(s). Once the association has been established, if no drop ports associated with an order entry port are connected, reject orders on the order entry port until at least one of the drop port session connections have been established. |
| Reject Orders on DROP Port Timeout (s) | 30 | Only applicable for sessions where “Reject Orders on Drop Port Disconnect” has been enabled. When the last associated drop port for the order entry session has disconnected, the reject/cancel actions will be taken on the order entry session if an associated drop port has not re-established its connection in the defined time. Minimum value allowed is 20. |
| Cancel Open Orders on Drop Port Disconnect | No | Only applicable for sessions where “Reject Orders on Drop Port Disconnect” has been enabled. If all drop ports associated with an order entry port become disconnected, cancel all open orders on the order entry port. |
| Port Order Rate Threshold | Default = 4000 msgs/sec<br><br>Maximum allowed = 4000 msgs/sec | The maximum allowed message rate on the session. When the first non-administrative message is received, a one second window begins. During the second no more than 3,999 additional non-administrative messages will be allowed within that window. If the rate is exceeded all new orders in the time window are rejected, modifies are treated as cancels, and cancels are processed. |
| Default Trading PID | none | The default trading PID to apply to messages from participants. |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

## 6. Support

Please direct questions or comments regarding this specification to TradeDeskAU@cboe.com.

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

# Appendix A – Login Strategies

### Login Message Flow

On a successful login, the following is the sequence of messages to expect. Note that after sending a `Login Request`, participants should wait for both a `Login Response` of "Accepted" as well as a `Replay Complete` message before sending new application messages.

| Direction | Message |
|---|---|
| Participant → Cboe Au | Login Request |
| Participant ← Cboe Au | Login Response (A = Accepted) |
| Participant ← Cboe Au | For each matching unit, all messages for the participant having sequence number greater than the seq # in the login request for that unit.<br>Possibly no such messages are transmitted, for instance on the first login of the day. |
| Participant ← Cboe Au | Replay Complete |
| Participant ←→ Cboe Au | (Begin normal flow: Order traffic, or Heartbeats at 1/sec) |

### Recommended Login

"Request replay of everything that you missed"

The login protocol supports a variety of options for how a participant may want to approach reconciling sequence numbers and messages after a disconnection. For most participants, the following Login strategy is the simplest approach to support a robust, lossless login experience:

| Field Name | Value | Comment |
|---|---|---|
| StartOfMessage | `BO E3` | |
| MessageLength | `1E 00` | 30 |
| MessageType | `01 00` | 1 (login request) |
| MatchingUnit | `00` | |
| Reserved | `00` | |
| SequenceNumber | `00` | (unsequenced) |
| SessionId | `41 42 43 44` | (ex. "ABCD") - Session Id as supplied by Cboe Australia |
| SessionSubId | `30 30 30 31` | (ex. "0001") Session Sub Id as supplied by Cboe Australia |
| Password | `61 62 63 64 65 66 67 68 69 6A` | (ex. "abcdefghij") Password |
| ReplayUnspecifiedUnit | `52` | "R" - Replay unknown units. |
| NumberOfUnits | `02` | There are 2 rows in the array below, representing the 2 system matching units. |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | |
|---|---|---|
| →UnitNumber | 01 | Matching unit #1 |
| →UnitSequence | 00 00 00 00 | Last SequenceNumber that participant received with MatchUnit=1. Zero if participant has not received anything for MatchUnit=1, for instance as the first login of the day. |
| →UnitNumber | 02 | Matching unit #2 |
| →UnitSequence | 00 00 00 00 | Last SequenceNumber that participant received with MatchUnit=1. Zero if participant has not received anything for MatchUnit=2, for instance as the first login of the day. |

**Other Login Strategies**

***No Replay***

"Just start with the next live event"

If the participant is confident that they are disinterested in any missed messages, they may elect to Skip (S) unspecified units and specify no units. This can be used as an initial login. Differences from the recommended login are in **Bold**.

| Field Name | Value | Comment |
|---|---|---|
| StartOfMessage | BO E3 | |
| MessageLength | 1E 00 | 30 |
| MessageType | 01 00 | 1 (login request) |
| MatchingUnit | 00 | |
| Reserved | 00 | |
| SequenceNumber | 00 | (unsequenced) |
| SessionId | 41 42 43 44 | (ex. "ABCD") - Session Id as supplied by Cboe Australia |
| SessionSubId | 30 30 30 31 | (ex. "0001") Session Sub Id as supplied by Cboe Australia |
| Password | 61 62 63 64 65 66 67 68 69 6A | (ex. "abcdefghij") Password |
| **ReplayUnspecifiedUnit** | 53 | **"S" – Skip replay of unspecified units.** |
| **NumberOfUnits** | 00 | **There are no rows in the array below.** |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

***Fail Login if any units are unspecified***

A minor variation on the "Recommended Login", this strategy will fail the login if not all known units are specified. This would provide some protection against a misconfiguration, or the unexpected introduction of additional matching units at Cboe Australia (although Cboe Australia would provide extensive notice and planning before introducing additional units). Differences from the recommended login are in **Bold**.

| Field Name | Value | Comment |
|---|---|---|
| StartOfMessage | BO E3 | |
| MessageLength | 1E 00 | 30 |
| MessageType | 01 00 | 1 (login request) |
| MatchingUnit | 00 | |
| Reserved | 00 | |
| SequenceNumber | 00 | (unsequenced) |
| SessionId | 41 42 43 44 | (ex. "ABCD") - Session Id as supplied by Cboe Australia |
| SessionSubId | 30 30 30 31 | (ex. "0001") Session Sub Id as supplied by Cboe Australia |
| Password | 61 62 63 64 65 66 67 68 69 6A | (ex. "abcdefghij") Password |
| **ReplayUnspecifiedUnit** | **46** | **"F" – Fail unknown units.** |
| NumberOfUnits | 02 | There are 2 rows in the array below, representing the 2 system matching units. |
| →UnitNumber | 01 | Matching unit #1 |
| →UnitSequence | 00 00 00 00 | Last SequenceNumber that participant received with MatchUnit=1. Zero if participant has not received anything for MatchUnit=1, for instance as the first login of the day. |
| →UnitNumber | 02 | Matching unit #2 |
| →UnitSequence | 00 00 00 00 | Last SequenceNumber that participant received with MatchUnit=1. Zero if participant has not received anything for MatchUnit=2, for instance as the first login of the day. |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

## Revision History

| Document Version | Date | Description |
|---|---|---|
| 1.0.0 | 03/05/22 | Initial draft. |
| 1.0.1 | 26/07/22 | Updated field position and offsets on multiple messages.<br>Corrected Symbol length and type in multiple messages.<br>Removed Capacity field from the Trade Cancel/Correct message.<br>Fixed text for “Cancel on Reject” port attribute to be specific to modify messages.<br>Corrected values for RestatementReason field (L=reload, W=would wash).<br>Updated MEQSE field to be for MOC and Pegs only (no limits). |
| 1.0.2 | 04/08/22 | Corrected field offset values for Cancel Order messages.<br>Corrected MessageLength and RiskGroupID offset for Purge Orders messages.<br>Corrected layout and table formatting of ShortSellNakedQty, ShortSellCoveredQty, BrokerPreferencing, and WorkingPrice for Order Acknowledgement messages.<br>Corrected RestatementReason references in LastPx and LastShare for Order Restated messages.<br>Updated RestatementReason value for "would wash" from R to W. |
| 1.0.3 | 15/08/22 | Corrected valid characters for Account field. |
| 1.0.4 | 26/08/22 | Updated SubLiquidityIndicator: removed code ‘m’, added codes I, J, K, S.<br>Updated example Login Request message in section 3.1.1.1. |
| 1.0.5 | 20/09/22 | Removed “not applicable for IOC, FOK” from MarketX.<br>Added references to equivalent FIX tags in Order Restated. |
| 1.0.6 | 14/10/22 | Corrected the Type values in Table 2 – Application Message Types. |
| 1.0.7 | 07/11/22 | Corrected RestatementReason code for iceberg reload from “L” to “r”. |
| 1.0.8 | 21/11/22 | Updated description of RiskGroupID.<br>Corrected description of MinQty when LeavesQty < MinQty.<br>Updated MessageLength description in sections 4.1.1. through 4.2.12.1, with the exception of section 4.1.4.1. |
| 1.0.9 | 10/01/23 | Corrected *IntermediaryID* data type to alphanumeric in section 4.3 |
| 1.0.10 | 31/01/23 | Remove “otherwise 0” from *Price* description since a cap price is required on pegged order.<br>Added *BaseLiquidityIndicator* and *SubLiquidityIndicator* to `Order Acknowledgement` message. |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | |
|---|---|---|
| 1.0.11 | 17/05/23 | Corrected specification language on valid characters for *Account*, *ClearingAccount*, and *IntermediaryID* fields. No underlying code changes.<br>Removed "Cancel Continuous Book Orders Only (default)" from "Cancel on Disconnect" port attribute, and changed default option to "Cancel All Open Orders (includes Market-On-Close orders)". |
| 1.0.12 | 24/01/24 | Clarified valid characters for *ClearingAccount* and updated data type to Alphanumeric. No underlying code changes. |
| 1.0.13 | 26/02/24 | Updated section 1.5 to include latency expectations as well as Members' responsibility to monitor the status of the messages they send to the exchange. |
| 1.0.14 | 30/04/24 | Addition of Post Only Orders, including *RoutingInst* and *DisplayIndicator* to:<br>• New Order Single V2<br>• Order Acknowledgement **V1**<br>• Port attributes.<br>Post-Only order updates are temporarily displayed in green font for clarity.<br><br>Addition of multi-day orders GTC and GTD with expiry date:<br>• New Order Single V2<br>• Purge Port<br>• Order Acknowledgement **V1**<br>• Port attributes<br>Multi-Day order updates are temporarily displayed in blue font for clarity.<br><br>Addition of Auction trade indicators:<br>• Order Execution message.<br>Auction updates are temporarily displayed in red font for clarity.<br>**Effective Date: Target late Q1, 2025 (tbc)** |
| 1.0.15 | 24/05/24 | Post Only Orders:<br>• Adjusted the *RoutingInst* field size to 1 character |
| 1.0.16 | 03/07/24 | Addition of four new messages for multi-day orders:<br>• ModifyOrderAUEquitiesV2 to allow clients to migrate from V1 to V2 overtime<br>• OrderModifiedAUEquitiesV2 to allow clients to migrate from V1 to V2 overtime<br>• CarriedRestatementAUEquitiesV1 to provide details of multi-day orders that were persisted overnight<br>• DoneForDayAUEquitiesV1 to provide details of multi-day orders that will be persisted at end of day<br><br>Support of multi-day GTD *ExpireDate* modification. |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved

| | | |
|---|---|---|
| 1.0.17 | 26/07/24 | Correction to include ***BaseLiquidityIndicator*** and ***SubLiquidityIndicator*** in 'OrderModified' messages, to align with actual system messages.<br><br>Further clarification relating to *ExpireDate* in the Modify Order section (removed the unrequired port attribute 'GTD date refresh on amend').<br><br>Further clarification in cancel-on-disconnect and purge scenarios around the separation between:<br>• Day orders, and<br>• GTC / GTD orders |
| 1.0.18 | 15/01/25 | Updated with Cboe Titanium branding. |
| 1.0.19 | 07/02/25 | Added *RestatementReason* = 'R' in `Order Restated` to accommodate orders amended by another port (i.e. Purge-Amend Port).<br>Purge-Amend Port updates are temporarily displayed in purple font for clarity.<br>**Effective Date: Target late Q1, 2025 (tbc)** |
| 1.0.20 | 22/05/26 | Added Focused Nearpoint option: *RoutingInst*=F (Focused Nearpoint which is only eligible to trade with opposing Farpoint). |

© 2026 Cboe Global Markets, Inc. or one or more of its affiliates.
All Rights Reserved